

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Certification of New Interstate	)	
Natural Gas Facilities	)	Docket No. PL18-1-000
	)	
Consideration of Greenhouse Gas	)	
Emissions in Natural Gas Infrastructure	)	Docket No. PL21-3-000
Project Reviews	)	
	)	

**INITIAL COMMENTS OF
THE EDISON ELECTRIC INSTITUTE**

I. INTRODUCTION

The Edison Electric Institute (“EEI”) respectfully submits the following comments in response to the Federal Energy Regulatory Commission’s (“Commission’s” or “FERC’s”) Draft Policy Statement on Consideration of New Interstates Natural Gas Facilities (“Draft Updated Certificate Policy Statement”)¹ and Draft Policy Statement on Consideration of Greenhouse Gas Emissions in Natural Gas Infrastructure Project Reviews (“Draft GHG Policy Statement”)² in the above-captioned docket. The Commission initially issued both policy statements on February 18, 2022. On March 24, 2022, the Commission issued an order designating the policy statements as drafts and setting an April 25, 2022 deadline for initial comments.³ EEI appreciates the opportunity to submit initial comments in response to the Draft Updated Certificate Policy Statement and the Draft GHG Policy Statement.

EEI is the association that represents all U.S. investor-owned electric companies. EEI

¹ *Consideration of New Interstate Natural Gas Facilities*, 178 FERC ¶ 61,107 (2022) (“Draft Updated Certificate Policy Statement”).

² *Consideration of Greenhouse Gas Emissions in Natural Gas Infrastructure Project Reviews*, 178 FERC ¶ 61,108 (2022) (“Draft GHG Policy Statement”).

³ *Order on Draft Policy Statements*, 178 FERC ¶ 61,197 (2022).

members provide electricity for more than 220 million Americans and operate in all 50 states and the District of Columbia. The electric power industry supports more than seven million jobs in communities across the United States. EEI's diverse membership includes electric utilities operating in all regions of the country, including traditional bilateral markets and markets operated by Regional Transmission Organizations ("RTOs") and Independent System Operators ("ISOs"), as well as combined gas-electric utilities.

EEI member companies invest more than \$120 billion annually to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure; to diversify the nation's energy mix; and to integrate new technologies that benefit both customers and the environment. EEI members are united in their commitment to get the energy they provide as clean as they can, as fast as they can, while keeping reliability and affordability front and center, as always, for the customers and communities they serve.

EEI members are in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation is being driven by a wide range of factors, including relatively lower prices for natural gas, particularly as compared to historic high prices, and renewable energy resources, energy efficiency and demand-side management, technological improvements, changing customer, investor and owner expectations, federal and state regulations and policies, and the increasing use of distributed energy resources. Across the nation, EEI member companies are leading a clean energy transformation, making significant progress to reduce greenhouse gas ("GHG") emissions in our sector, while also creating good-paying jobs and an equitable clean energy future. With the right policies and technologies, a 100 percent clean energy future can be more than a goal, it can be a reality. Across the industry, companies are investing in a broad range of affordable, carbon-free technologies and approaches

with the goal of finding the most cost-effective ways to deliver resilient clean energy.

Advancing these technologies from idea to implementation requires time and investment. EEI and its members support policies that provide dramatic increases in funding for the research, development, demonstration, and deployment of these technologies.

As discussed in greater detail below, natural gas-based generation has played—and will continue to play—an essential role in the electric power sector’s transformation. Interstate natural gas pipeline infrastructure supports the electric power sector’s reliable transition to lower emissions fuel sources. As a result, transparency and regulatory certainty in Commission processes related to Natural Gas Act (“NGA”) jurisdictional facilities, as well as ensuring that its NGA authorizations are able to withstand judicial review, are critical. In order to support its goal of providing “greater certainty and predictability for all stakeholders,”⁴ any action that the Commission takes in the instant proceeding and in implementing the Draft Updated Certificate Policy Statement and Draft GHG Policy Statement must be consistent with its authority under the NGA and the procedural obligations imposed by the National Environmental Policy Act (“NEPA”). The Commission’s GHG analyses under NEPA should be robust and provide proper contextualization of emissions to allow for a complete assessment of potential emissions impacts from proposed infrastructure. Furthermore, other federal agencies, including the U.S. Environmental Protection Agency (“EPA”) and the Pipeline and Hazardous Materials Safety Administration (“PHMSA”), have jurisdiction over issues related to natural gas pipeline emissions. The Commission should defer to these agencies where they are expert and have jurisdiction and should coordinate with its sister agencies to avoid duplicating efforts and to support administrative efficiency and avoid imposing conflicting or incompatible requirements.

⁴ Draft GHG Policy Statement at P 1.

II. COMMENTS

A. Electric Companies Continue to Lead the Clean Energy Transition and Natural Gas is Expected to Continue to Play a Significant Role in Reducing Power Sector Emissions

The mix of resources used to generate electricity in the United States has changed dramatically over the last decade and is increasingly cleaner. 2016 marked the first year that natural gas surpassed coal as the main source of electricity generation in the United States. In 2020 natural gas-based generation powered about 38 percent of the country's electricity, compared to coal-based generation at about 22 percent.⁵ Renewables⁶ generated approximately 21 percent of total generation.⁷ In total, approximately 40 percent of America's electricity came from carbon-free resources in 2021, including nuclear energy, hydropower, solar, and wind.⁸

Renewable energy deployments will continue. By 2025, EIA projects approximately 125 GW of renewables capacity will be online.⁹ Further, EIA projects that in the United States the share of renewables in the electricity generation mix will more than double by 2050.¹⁰ EIA

⁵ See Energy Information Administration (EIA), *Electric Power Monthly: with Data for December 2021*, 12 (Feb. 2022), https://www.eia.gov/electricity/monthly/current_month/february2022.pdf.

⁶ Renewables here are defined as wind, hydroelectric, solar, biomass, and geothermal energy.

⁷ See *supra* note 5.

⁸ See *id.*

⁹ See EIA, *Annual Energy Outlook 2022: Reference Case Projections Tables – Table 16. Renewable Energy Generating Capacity and Generation* (Mar. 3, 2022), <https://www.eia.gov/outlooks/aeo/data/browser/#/?id=16-AEO2022&cases=ref2022&sourcekey=0>.

¹⁰ See EIA, *Annual Energy Outlook 2022: With Projections To 2050 – Narrative 17* (Mar. 3, 2022), https://www.eia.gov/outlooks/aeo/pdf/AEO2022_Narrative.pdf. In addition, EIA notes that coal use will continue to decline with the retirement of most of the relatively old and inefficient coal-fired electricity generating units in the United States. *Id.* at 18. EIA estimates are intentionally conservative, focusing on policies currently on the books and not other potential drivers of increased renewable energy deployment, including a suite of clean energy tax credits currently being considered by Congress. These credits will drive reductions in the costs of a range of clean energy sources, increasing both deployment and emissions reductions relative to the EIA base case. See, e.g., Michael Greenstone, et al., *Assessing the Costs and Benefits of Clean Electricity Tax Credits, Build Back Better Act Policy Memo*, Energy Policy Institute, University of Chicago, and Rhodium Group (Feb. 9, 2022), <https://rhg.com/research/assessing-the-costs-and-benefits-of-clean-electricity-tax-credits/#:~:text=Building%20on%20previous%20modeling%20conducted,a%20scenario%20without%20these%20policies>.

projects that wind will continue to be responsible for most of the growth in renewables generation through 2024, accounting for more than two-thirds of those increases in electricity generation during that period and that solar will dominate deployments thereafter until 2050.¹¹

These changes have had a profound impact on the sector's carbon dioxide ("CO₂") emissions, the primary GHG emissions associated with electricity production. We are proud to provide America's resilient clean energy and to be leading the transformation of energy. Preliminary full-year estimates are that electric power sector emissions were 36 percent below 2005 levels as of the end of 2021, as low as they were in 1984.¹² These reductions will continue. Fifty EEI members have announced forward-looking carbon reduction goals, more than two-thirds of which include a net-zero by 2050 or earlier equivalent goal, and members are routinely increasing the ambition or speed of their goals or altogether transforming them into net-zero goals.

In addition, the electric industry has significantly reduced air pollutants such as mercury, hazardous air pollutants, sulfur dioxide ("SO₂"), and nitrogen oxides ("NO_x"). As of 2021, SO₂ and NO_x emissions have declined 94 and 88 percent, respectively, since 1990.¹³ In addition, mercury emissions have declined by 90 percent since 2010,¹⁴ and total HAPs—including all acid gas emissions—declined by 96 percent between 2010 to 2017.¹⁵

New and existing natural gas-based electric generating units are expected to continue to play a significant role in reducing emissions from the power sector while helping to keep

¹¹ See *id.*

¹² See EIA, Monthly Energy Review, Table 11.16—Electric Power Sector (Mar. 29, 2022), <https://www.eia.gov/totalenergy/data/monthly/>.

¹³ See EPA, EPA Issues Power Plant Emissions Data for 2021 (Feb. 22, 2022), <https://www.epa.gov/newsreleases/epa-issues-power-plant-emissions-data-2021>.

¹⁴ See *id.*; EPA, *Regulatory Impact Analysis for the Final Mercury and Air Toxics Standards*, 2-7 (Dec. 2011), <https://www.epa.gov/sites/default/files/2015-11/documents/matsriafinal.pdf>.

¹⁵ See 84 *Fed. Reg.* 2,670, 2,689 (Feb. 7, 2019).

electricity affordable and reliable for customers. In fact, recent studies examining pathways to achieve a net-zero economy find that there are more natural gas-based combustion turbines deployed in 2050 than there are today.¹⁶ That is because these units, at this time, are still the most cost-effective flexible generation capable of responding to variable renewable energy production. This future turbine fleet may have lower capacity factors than today, may burn natural gas but use carbon capture and storage, or may use low- or no-carbon fuels, like hydrogen and ammonia.

EEI supports continued methane emissions reductions attributable to the upstream natural gas supply chain—minimizing upstream methane emissions is critical to the industry’s ability to continue to utilize natural gas. While natural gas-based power generation emits significantly less carbon than coal-based generation, it is essential that methane emissions from the production and transportation of natural gas do not offset some of that benefit. Maximizing supply chain efficiencies, including minimizing methane emissions from the upstream natural gas supply chain operations and infrastructure is essential to the industry’s ability to continue to utilize natural gas.

In the long term, reaching net-zero carbon emissions also will require the deployment of next-generation, carbon-free, 24/7, dispatchable technologies not currently available commercially. Developing a broad range of advanced clean energy technologies can help expedite the transition of the electric power sector to one that is low- or non-emitting while keeping electricity affordable and reliable for customers. EEI members remain committed to

¹⁶ See, e.g., James H. Williams, et al., *Carbon Neutral Pathways for the United States*, 11-12, AGUAdvances, 2 (Nov. 2020), <https://doi.org/10.1029/2020AV000284>; see also Eric Larson, et al., *Net Zero America: Potential Pathways, Infrastructure and Impacts*, Final Report, 150 (Oct. 29, 2021), <https://netzeroamerica.princeton.edu/the-report>.

providing customers with affordable and reliable power, and a successful decarbonization strategy must ensure that Americans have access to reliable, safe, and reasonably priced electricity.

B. In Undertaking Its NEPA Analyses the Commission Should Do Robust Analyses and Place Those Analyses in Context

EEI has previously noted in comments to the Commission that discussion of GHG impacts in the context of pipeline certification is best approached through the existing NEPA framework and related precedent.¹⁷ The Council on Environmental Quality’s (“CEQ’s”) regulations implementing NEPA define “effects” as those “changes to the human environment from the proposed action or alternatives that are reasonably foreseeable and have a reasonably close causal relationship to the proposed action or alternatives.”¹⁸ The U.S. Court of Appeals for the District of Columbia Circuit (“D.C. Circuit”) has found that downstream emissions related to the Commission’s authorization of interstate natural gas pipeline facilities may qualify as reasonably foreseeable effects for NEPA purposes.¹⁹

¹⁷ *EEI Post-Technical Conference Comments*, Jan. 7, 2022 (“EEI Post-Technical Conference Comments”); and *EEI 2018 Certificate Policy Statement Comments*, July 25, 2018 (“EEI Certificate Policy Statement Comments”).

¹⁸ 40 C.F.R. § 1508.1(g).

¹⁹ In *Food & Water Watch v. FERC*, the D.C. Circuit explained that its “precedents establish that downstream emissions are not, ‘as a categorical matter, always a reasonably foreseeably indirect effect of a pipeline project.’ Rather, foreseeability depends on information about the ‘destination and end use of the gas in question.’” 28 F.4th 277, 288 (D.C. Cir. 2022) (citing *Birckhead v. FERC*, 925 F.3d 510, 519 (D.C. Cir. 2019)). Further, the D.C. Circuit has found that a case-by-case inquiry is required to determine whether emissions from downstream natural gas combustion are a reasonably foreseeable effect of the Commission’s authorization of a particular project. *Birckhead*, 925 F.3d at 519 (citing *Calvert Cliffs’ Coordinating Committee, Inc. v. U.S. Atomic Energy Commission*, 449 F.2d 1109, 1122 (D.C. Cir. 1971) (“NEPA compels a case-by-case examination . . . of discrete factors.”)). For example, in *Sabal Trail v. FERC*, the D.C. Circuit found that emissions from downstream power plants associated with the Commission’s authorization of an interstate natural gas pipeline project were reasonably foreseeable effects under NEPA where the pipeline was designed to deliver natural gas to power plants that had been specifically identified. 867 F.3d 1357, 1371-73 (D.C. Cir. 2017)). By contrast, in *Birckhead*, the Court “rejected a similar indirect-effects claim when the Commission could establish only that ‘the gas [was] headed somewhere in the Southeast.’” *Food & Water Watch*, 28 F.4th at 288 (citing *Birckhead*, 925 F.3d at 518). Most recently, in *Food & Water Watch*, the D.C. Circuit concluded that the end use of the gas transported was reasonably foreseeable where the Commission “knew, with a good deal of specificity, where the gas in question would be going (to Columbia Gas’s existing customers in the Greater Springfield area) and how it would be used (to fuel residential and commercial gas connections).” *Id.*

The Commission’s Draft GHG Policy Statement suggests an approach that will conform to the scope of its NEPA responsibilities and recent D.C. Circuit rulings.²⁰ It is important that the Commission address the D.C. Circuit’s perspective as it applies this policy statement on a case-by-case basis in order to ensure that the Commission’s decisions withstand judicial review and that the natural gas infrastructure needed to support the electric sector’s ongoing clean energy transformation is available.

NEPA requires federal agencies to consider environmental impacts of proposed actions as part of agencies’ decision-making processes. NEPA is not, in and of itself, determinative of project approval as it simply prescribes the steps by which agencies must evaluate options and explain their decision-making. Thus, while GHG emissions alone cannot and should not drive federal NEPA decision-making, agencies should take steps to reasonably quantify reasonably foreseeable GHG emissions—and appropriately contextualize and caveat such quantifications—to satisfy NEPA’s informational requirements and to minimize challenges to environmental reviews on the grounds that they did not adequately consider such emissions.

Other environmental, contextual, and statutory factors may dictate that more detailed GHG emissions analyses be undertaken, particularly if they help to put emissions in the appropriate context and may help projects withstand judicial delays. In the context of some natural gas pipelines, the downstream use of natural gas in electricity generation could be part of an effort to reduce emissions from other fossil-based generation and to support the integration of an increasing amount of variable renewable generation, thereby reducing overall sector emissions. In addition, downstream use of natural gas in electricity generation, including natural gas generation in the absence of or in conjunction with carbon-free technologies, may be

²⁰ Specifically, the Commission notes in several instances throughout the Draft GHG Policy Statement that it proposes to consider, on a case-by-case basis, whether upstream and downstream emissions are a reasonably foreseeable effect of an NGA section 7 project. *See, e.g.*, Draft GHG Policy Statement at PP 31, 38, and 43.

necessary to meet the ongoing need for dispatchable generation technologies to integrate the ever-growing nationwide fleet of renewable generation in a reliable and cost-effective manner.

The Commission's GHG analyses should acknowledge these factors.

A complete, contextual analysis serves two goals: 1) it provides the information necessary for an informed analysis of environmental impacts as required by NEPA; and 2) it will help produce legally sufficient NEPA analyses that will withstand D.C. Circuit review, thus avoiding costly litigation delays in the permitting of critical energy infrastructure that cause investor uncertainty and divert resources from clean energy projects.

EEI supports the Commission's inclusive approach to quantification of a project's reasonably foreseeable emissions, including consideration of record evidence of net-reductions in GHG emissions where the use of the natural gas to be transported on the NGA section 7 facilities would displace the use of higher emitting fuels²¹ and where such facilities would be used to transport emerging low- or no-emissions fuels, including hydrogen.²² Sector-wide contextualization is also important, particularly when agencies undertake NEPA GHG analyses for specific energy projects. When undertaking its NEPA analysis, the Commission should consider power sector reductions in GHG emissions for projects or proposed actions that implicate electricity generation when reasonable and suggested by the project proponent. Sector-wide assessments of total GHG emissions and emissions trends can help the Commission and interested stakeholders better understand the potential direct, indirect, and cumulative emissions impacts of a proposed action. For example, sector-wide assessments can help demonstrate that a

²¹ Draft GHG Policy Statement at P 52.

²² As noted in Section II.A above, future electric generation turbines may use low- to no-carbon fuels, including hydrogen and ammonia. Multiple industry stakeholders, including the Department of Energy and PHMSA, are exploring the ability to use NGA section 7 infrastructure to facilitate transportation of emerging fuels. The ability to do so could accelerate deployment and reduce overall costs for the emerging fuels.

particular project—such as a natural gas pipeline—serves to reduce indirect emissions by allowing electric companies to integrate more renewable resources or shift fuel sources.

Where GHG emissions are difficult to quantify, sector-wide emissions assessments can provide helpful context that allows agencies to provide a qualitative assessment of the potential impacts of a proposed permit or action. Reliable sector-wide assessments are readily available and can help inform both agency decision-making and interested stakeholders consistent with the requirements and objectives of NEPA.

Further, EEI broadly supports efforts to reduce and mitigate upstream methane emissions—ensuring that methane emissions are appropriately mitigated is essential to the industry’s ongoing transformation.²³ To ensure a legally sufficient analysis that will withstand D.C. Circuit review, the Commission should make certain that it is requiring project developers to provide information in line with reasonably identifiable and quantifiable methods used throughout the industry today. Several panelists in the Commission’s November 19, 2021 technical conference in the instant proceeding raised concerns around the current ability to identify and quantify pipeline emissions.²⁴ In comments responding to CEQ’s proposals

²³ EEI’s members were instrumental in organizing the Natural Gas Sustainability Initiative (“NGSI”), a voluntary, industry-led effort to advance best practices and to encourage continuous improvement in methane emissions reductions through company-level disclosure. NGSI is intended to complement and work in concert with regulatory standards from the EPA, which are critical to reducing emissions and providing certainty to both the regulated industry and its customers in the supply chain. NGSI also aims to provide consistency and comparability to these measurements throughout the supply chain and to encourage continuous improvement in methane emissions reductions through company-level disclosure.

²⁴ See *Technical Conference on Greenhouse Gas Mitigation: Natural Gas Act Sections 3 and 7 Authorizations*, Dkt. No. PL21-3-000, Nov. 19 Tr. at 189 (Humes). Notably, PHMSA held a public meeting and forum in late 2021 addressing various pipeline safety topics, including “solutions to reduce methane releases from PHMSA regulated infrastructure.” *Meetings and Documents, Pipeline Transportation: Hydrogen and Emerging Fuels R&D Public Meeting and Forum*, PIPELINE AND HAZARDOUS MATERIALS SAFETY ADMIN., <https://primis.phmsa.dot.gov/meetings/MtgHome.mtg?mtg=153>. PHMSA convened work groups during the meeting to discuss various topics related to its pipeline safety regulations. Work group 5 was assigned Methane Mitigation – Construction and Operations. *Id.* In its December 2, 2021 report, this work group identified research gaps that are particularly germane to the instant proceeding, including the need for further research into “technologies to detect, accurately locate, and quantify methane emissions in diverse operating areas and environments.” *Work Group # 5: Methane Mitigation – Construction and Operations, Pipeline Transportation: Hydrogen and Emerging Fuels R&D*

addressing GHG emissions in NEPA analyses, EEI has encouraged CEQ to balance NEPA’s information disclosure requirements with the inherent difficulties of quantifying reasonably foreseeable direct and indirect GHG emissions related to a specific proposed action or project.²⁵

C. The Commission Must Protect the Ability of Its Decisions to Withstand Judicial Review

The Commission must carefully analyze its statutory authority to take the action proposed in the Draft Updated Certificate Policy Statement and the Draft GHG Policy Statement, since the NGA itself does not provide the Commission with any direct or enumerated authority to regulate environmental performance.²⁶ While some environmental issues may factor into the Commission’s consideration of benefits and adverse effects in its NGA Section 7 balancing test, the Commission’s analysis of environmental impacts derives from its responsibilities under NEPA.

As noted above, NEPA is a fundamentally procedural statute that “merely prohibits uninformed . . . agency action.”²⁷ NEPA’s action-forcing procedures require agencies to take a hard look at potential environmental impacts and “provide for broad dissemination of relevant environmental information.”²⁸ As the U.S. Supreme Court has explained, “it is now well settled that NEPA itself does not mandate particular results, but simply prescribes the necessary

Public Meeting and Forum, November 30-December 2, 2021, PIPELINE AND HAZARDOUS MATERIALS SAFETY ADMIN., at pp 2-3, <https://primis.phmsa.dot.gov/meetings/FilGet.mtg?fil=1215>.

²⁵ See *Comments from the Edison Electric Institute on the Council on Environmental Quality’s Notice of Proposed Rulemaking National Environmental Policy Act Implementation Regulations Revisions*, CEQ-2021-0002 (Nov. 22, 2021), attached hereto as Attachment A.

²⁶ Those authorities have been delegated by Congress in other environmental statutes to, primarily and understandably, the EPA—the nation’s expert Agency in environmental regulation.

²⁷ *Robertson v. Methow Valley Citizens Council*, 490 U.S. 332, 351 (1989) (“Other statutes may impose substantive environmental obligations on federal agencies, but NEPA merely prohibits uninformed . . . agency action.”).

²⁸ *Id.* at 349.

process.”²⁹ Furthermore, while CEQ’s regulations implementing NEPA require agencies to discuss possible mitigation in NEPA reviews, “NEPA does not impose a substantive duty on agencies to mitigate adverse environmental effects”³⁰ nor “a substantive requirement that a complete mitigation plan be actually formulated and adopted.”³¹ In sum, NEPA is a fundamentally procedural requirement that provides a current snapshot of potential environmental impacts and potential mitigation options, but does not provide broad authority to impose those mitigation measures.

Accordingly, as EEI previously has noted,³² an assessment of GHG emissions (including any information provided by an applicant) and potential impacts under NEPA should inform the Commission’s evaluations, but such analysis does not necessarily dictate the ultimate outcome of the larger balancing test that the Commission undertakes consistent with the NGA’s requirements. To support the ability of its decisions to withstand judicial review, the Commission must make certain that its efforts to analyze emissions and identify potential mitigation measures for emissions rest within its statutory authority³³ and do not conflict with the statutory authority of other expert federal agencies that have a primary responsibility for

²⁹ *Id.* (citing *Strycker’s Bay Neighborhood Council, Inc. v. Karlen*, 444 U.S. 223, 223-28 (1980); and *Vermont Yankee Nuclear Power Corp. v. Natural Resources Defense Council, Inc.*, 435 U.S. 519, 558 (1978)).

³⁰ *Id.* at 333.

³¹ *Id.* at 352.

³² See EEI Post-Technical Conference Comments at 8; and EEI Certificate Policy Statement Comments at 10.

³³ While NGA section 7 permits the Commission to attach “such reasonable terms and conditions as the public convenience and necessity may require,” 15 U.S.C. §717f(e), a mitigation measure must not amount to a *de facto* denial. See Draft GHG Policy Statement (Christie, Comm’r, dissenting at n.55) (citing *Nat’l Fuel Gas Supply Corp. v. FERC*, 909 F.2d 1519, 1522 (D.C. Cir. 1990) (“The Commission may not, . . . when it lacks the power to promote the public interest directly, do so indirectly by attaching a condition to a certificate that is, in its unconditional form, already in the public convenience and necessity.”) (citations omitted in original))). Furthermore, while EEI supports the Commission’s inclusive approach to quantifying reasonably foreseeable GHG emissions, as discussed in Section II.B *supra*, the Commission must ensure that any requirements imposed in its orders are limited to those entities over which the Commission has jurisdiction under Section 7 of the NGA. In addition, the Commission should consider adopting measures to ensure that any mitigation requirements or analyses remain focused on the certificate process.

environmental regulation. Ultimately, these concerns can be addressed by adopting a holistic approach to GHG analysis under NEPA, where placing those emissions in context can help buttress the Commission’s approach to both NEPA analysis and its obligations under NGA Section 7.

In addition, while EEI supports the Commission’s flexible approach in the Draft Updated Certificate Policy Statement and Draft GHG Policy Statement, the Commission must apply these policy statements in a reasonably consistent manner as it considers applications on a case-by-case basis. Uniform and consistent application would provide applicants certainty while decreasing the risk of litigation that could further delay or impede deployment of infrastructure that is important to the electric sector’s ability to continue its clean energy transformation. FERC must also continue efforts to maintain an efficient review process and ensure that its decisions remain reasoned and grounded in reliable evidence, even as it considers a potentially broader range of information.

D. The Commission Should Collaborate with Other Agencies and Defer to EPA on the Sufficiency of GHG Mitigation Measures

The Commission should continue to work collaboratively with its sister agencies on GHG emissions issues within the scope of its jurisdiction, including EPA and PHMSA.³⁴ Any inconsistencies between and among the Commission’s attempts to mitigate GHG emissions and other federal agencies’ exercise of their respective jurisdiction could create confusion for project developers, end-users, and the investment community.³⁵

³⁴ Such efforts also would align with Executive Order 14008, which called on federal agencies to take a coordinated approach to meeting the challenges of climate change. Exec. Order. No. 14008, 86 *Fed. Reg.* 7612, 7622 and 7628 (Feb. 1, 2021).

³⁵ EEI notes the Commission’s statement that “CEQ and EPA are undertaking initiatives that may culminate in the establishment of a significance threshold for GHG emissions or that may further impact the Commission’s determination of GHG significance in its NEPA analysis. If CEQ or EPA issues any future guidance regarding the

While the Commission’s NEPA obligations require it to consider environmental impacts of any proposed action that significantly affects the environment, the Commission should defer to the EPA with respect to environmental risk analysis and the sufficiency of emissions mitigation measures. Congress, through the Clean Air Act (“CAA”), provided EPA with jurisdiction over air quality and the reduction and control of air pollution nationwide. EPA has exercised its CAA authority over various parts of the energy sector, including its recent laudable efforts to regulate direct emissions from the natural gas industry using CAA section 111.³⁶

Given EPA’s expertise, its jurisdiction under the CAA, and its recent and ongoing actions specifically related to emissions in the natural gas sector, the Commission should defer to EPA on the sufficiency of emissions mitigation measures.³⁷ Such deference would avoid the potential for inconsistent requirements between the agencies and comport with the Commission’s existing NEPA process for review of NGA-jurisdictional facilities, in which EPA already participates.

II. CONCLUSION

EEI appreciates the opportunity to submit initial comments on the Draft Updated Certificate Policy Statement and Draft GHG Policy Statement and looks forward to continuing to participate in the discussion.

evaluation of GHG emissions, the Commission may adjust its methods for determining the significance of GHG emissions consistent with that guidance.” Draft GHG Policy Statement at n.206.

³⁶ On November 15, 2021, EPA published a proposed rule in the Federal Register that “comprises three distinct groups of actions under the [CAA] which are collectively intended to significantly reduce emissions of [GHGs] and other harmful air pollutants from the . . . Natural Gas source category.” Standards of Performance for New, Reconstructed, and Modified Sources and Emissions Guidelines for Existing Sources: Oil and Natural Gas Sector Climate Review, 86 *Fed. Reg.* 63,110 (Nov. 15, 2021). The Federal Register notice explains that the Natural Gas source category includes “natural gas production, processing, transmission, and storage” and that “the focus is on all operations from the well to the local distribution company custody transfer station commonly referred to as the ‘city-gate.’” *Id.* at 63,113 n.2.

³⁷ The Commission should also add existing state-based carbon policies, including those administered by the California Air Resources Board, the Regional Greenhouse Gas Initiative, the Colorado Energy Plan, and state renewable portfolio standards, to the list mitigation mechanisms that project sponsors may. Draft GHG Policy Statement at PP 113-28.

Respectfully Submitted,

/s/ *Emily Fisher*

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April 25, 2022

ATTACHMENT A



Edison Electric
INSTITUTE

Quinlan J. Shea, III

Vice President, Environment, Natural Resources,
and Occupational Safety & Health

November 22, 2021

Brenda Mallory
Chair
Council on Environmental Quality
730 Jackson Place NW
Washington, DC 20503

RE: Docket No. CEQ-2021-0002

[Submitted Electronically]

Dear Chair Mallory:

The Edison Electric Institute (EEI) appreciates the opportunity to submit comments to the Council on Environmental Quality (CEQ) on the notice of proposed rulemaking *National Environmental Policy Act Implementation Regulations Revisions* (NOPR or Proposal). 86 *Fed. Reg.* 55,757 (Oct. 7, 2021). CEQ is proposing to update the National Environmental Policy Act (NEPA) implementing regulations as they relate to the purpose and need statement and the scope of reasonable alternatives, agency NEPA procedures, and the definition of “effects.” On all three issues, the NOPR would rescind the current regulations in place and revert to the definitions that were in place prior to the 2020 revisions.

EEI is the association that represents all U.S. investor-owned electric companies. EEI members provide electricity for more than 220 million Americans and operate in all 50 states and the District of Columbia. As a whole, the electric power industry supports more than seven million jobs in communities across the United States. EEI member companies invest more than \$140 billion annually to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure; to diversify the nation’s energy mix; and to integrate new technologies that benefit customers. EEI members are united in their commitment to get the energy they provide as clean as they can, as fast as they can, while keeping reliability and affordability front and center, as always, for the customers and communities they serve. Across the nation, EEI member companies are leading a clean energy transformation, making significant progress to reduce greenhouse gas (GHG) emissions in our sector, while also creating good-paying jobs and an equitable clean energy future.

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EEl member companies' critical infrastructure projects frequently require federal permits and, therefore, EEl members support revisions to the NEPA implementing regulations that facilitate their ability to build and maintain critical infrastructure as they deploy increasing amounts of clean energy. EEl's comments focus on EEl members' clean energy transformation and the resultant GHG emissions reductions; the appropriateness of considering the applicant's purposes when considering the purpose of and need for a project as well as consideration of reasonable alternatives; the desire for consistency across multiple agencies as they apply their own NEPA regulations; and the importance of agencies taking steps to quantify GHG emissions reasonably—and appropriately contextualize and caveat such quantifications—to satisfy NEPA's informational requirements as well as minimize challenges to environmental reviews on the grounds that they did not adequately address such emissions.

Thank you for your attention to these comments, questions on these comments may be directed to [Alex Bond](#) (202-508-5523), [Eric Holdsworth](#) (202-508-5103) or [Riaz Mohammed](#) (202-508-5036).

Sincerely,

A handwritten signature in black ink, appearing to read "Quinlan J. Shea, III". The signature is fluid and cursive, with a prominent "Q" and "S".

Quinlan J. Shea, III

**COMMENTS FROM THE EDISON ELECTRIC INSTITUTE ON THE COUNCIL ON
ENVIRONMENTAL QUALITY'S NOTICE OF PROPOSED RULEMAKING
NATIONAL ENVIRONMENTAL POLICY ACT IMPLEMENTATION REGULATIONS
REVISIONS**

Docket No. CEQ-2021-0002

November 22, 2021

The Edison Electric Institute (EEI) appreciates the opportunity to submit comments to the Council on Environmental Quality (CEQ) on the notice of proposed rulemaking *National Environmental Policy Act Implementation Regulations Revisions* (NOPR or Proposal). 86 *Fed. Reg.* 55,757 (Oct. 7, 2021). CEQ is proposing to update the National Environmental Policy Act (NEPA) implementing regulations as they relate to the purpose and need statement and the scope of reasonable alternatives, agency NEPA procedures, and the definition of “effects.” On all three issues, the NOPR would rescind the current regulations in place and revert to the definitions that were in place prior to the 2020 revisions.

EEI member companies invest more than \$140 billion annually to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure in order to provide affordable and reliable electricity to customers. Such critical infrastructure projects frequently require federal permits and, therefore, EEI members support revisions to the NEPA implementing regulations that support their ability to build and maintain critical infrastructure as they deploy increasing amounts of clean energy.

I. Introduction and Executive Summary.

EEI is the association that represents all U.S. investor-owned electric companies. EEI members provide electricity for more than 220 million Americans and operate in all 50 states and the District of Columbia. As a whole, the electric power industry supports more than seven million

jobs in communities across the United States. EEI member companies invest more than \$140 billion annually to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure; to diversify the nation's energy mix; and to integrate new technologies that benefit customers. EEI members are united in their commitment to get the energy they provide as clean as they can, as fast as they can, while keeping reliability and affordability front and center, as always, for the customers and communities they serve. Across the nation, EEI member companies are leading a clean energy transformation, making significant progress to reduce greenhouse gas (GHG) emissions in our sector, while also creating good-paying jobs and an equitable clean energy future.

These comments focus on EEI members' clean energy transformation and the resultant GHG emissions reductions; the appropriateness of considering the applicant's purposes when considering the purpose of and need for a project as well as consideration of reasonable alternatives; the desire for consistency across multiple agencies as they apply their own NEPA regulations; and the importance of agencies taking steps to quantify GHG emissions reasonably—and appropriately contextualize and caveat such quantifications—to satisfy NEPA's informational requirements as well as minimize challenges to environmental reviews on the grounds that they did not adequately address such emissions. EEI looks forward to engaging with CEQ on these and future potential revisions to the NEPA regulations to meet our clean energy goals.

II. Electric Companies Continue to Lead the Clean Energy Transformation.

EEI members are in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation is being driven by a wide range of factors,

including declining costs for natural gas and renewable energy resources, energy efficiency and demand-side management, technological improvements, changing customer expectations, federal and state regulations and policies, and the increasing use of distributed energy resources. EEI members are well-positioned to lead the way toward a climate solution. With the right policies and technologies, a 100-percent clean energy future can be more than a goal. It can be a reality. Across the industry, companies are investing in a broad range of affordable, carbon-free technologies and approaches with the goal of finding the most cost-effective ways to deliver clean, reliable energy. Advancing these technologies from idea to implementation requires time and investment. We support policies that provide dramatic increases in funding for the research, development, demonstration, and deployment of these technologies.

The mix of resources used to generate electricity in the United States has changed dramatically over the last decade and is increasingly cleaner. Starting in 2016, natural gas surpassed coal as the main source of electricity generation in the United States, and in 2020 natural gas-based generation powered 40 percent of the country's electricity, compared to coal-based generation at 19 percent.¹ Over the past eight years, more than half of the industry's investments in new electricity generation have been in wind and solar generation resources,² and 40 percent of

¹ See Energy Information Administration (EIA), *Electric Power Monthly* (Feb. 2021), <https://www.eia.gov/electricity/monthly/>. According to data in the same EIA report, in 2020, U.S. annual electricity generation from renewable sources exceeded coal-based generation, the first time that has occurred on an annual basis. Renewable sources provided 20.5 percent of total electricity generation in comparison to coal's 19 percent.

² See EIA, *Nearly Half of Utility-Scale Capacity Installed in 2017 Came from Renewables* (Jan. 10, 2018), <https://www.eia.gov/todayinenergy/detail.php?id=34472>; see also EEI, Industry Data, Statistical Highlights: Capacity and Generation (2018), <http://www.eei.org/resourcesandmedia/industrydataanalysis/industrydata/Pages/default.aspx>.

America's electricity in 2020 was generated from carbon-free resources, including nuclear energy, hydropower, solar, and wind.³ Notably, electric companies provide 67 percent of the solar energy in the U.S.

The trend of increasing renewable energy deployments will continue: EIA projects that in the United States the share of renewables in the electricity generation mix will more than double by 2050.⁴ Wind will be responsible for most of the growth in renewables generation through 2024, accounting for more than two-thirds of those increases in electricity generation during that period.⁵ Domestically, the long-term demand for new electric generating capacity will be met by renewables and efficient natural gas as older coal-based and less-efficient natural gas-based generating units retire. EIA notes that coal use will generally decline through 2050 with the retirement of coal-fired electricity generating units in the United States.⁶

These changes have had a profound impact on the sector's carbon dioxide emissions, the primary GHG emissions associated with electricity production. Overall, based on preliminary estimates from EIA, electric power sector emissions were 40 percent below 2005 levels as of the end of 2020, their lowest level in more than 40 years.⁷ These reductions will continue: Four dozen EEI members have announced forward-looking carbon reduction goals, more than half of which

³ See EIA, n.1, *supra*.

⁴ See EIA, *Annual Energy Outlook 2021: With Projections to 2050*, at 16 (Feb. 2021), https://www.eia.gov/outlooks/aeo/pdf/AEO_Narrative_2021.pdf.

⁵ See *id.* at 7-8.

⁶ *Id.* at 8.

⁷ EIA, *Monthly Energy Review* (Mar. 2021), <https://www.eia.gov/totalenergy/data/monthly/>.

include a net-zero by 2050 or earlier equivalent goal. EEI's member companies see a clear path to continued emissions reductions over the next decade using current technologies, including nuclear power, natural gas-based generation, energy efficiency, energy storage, and deployment of new renewable energy, especially wind and solar. These technologies will enable significant, cost-effective carbon reductions.

In the long term, achieving the deep reductions that are needed to get to net-zero carbon emissions will require the deployment of next-generation, carbon-free, 24/7, dispatchable technologies not currently available commercially. Developing a broad range of advanced clean energy technologies can help expedite the transition of the electric power sector to one that is low- or non-emitting while keeping electricity affordable and reliable for customers. EEI members remain committed to providing customers with affordable and reliable power, and a successful decarbonization strategy must ensure that Americans have access to reliable, safe, and reasonably priced electricity. EEI members are also committed to working with policymakers to advance policies that will achieve breakthroughs in making carbon-free, 24/7, dispatchable technologies commercially available and affordable so electric companies can achieve their long-term carbon reduction goals and support economywide long-term reductions.

An efficient, effective, and balanced NEPA process will allow EEI members to continue the clean energy transformation while also ensuring that agencies take a "hard look" at environmental consequences as mandated by NEPA.

III. CEQ Should Account for the Applicant's Goals and Objectives as Part of a Holistic Review of the Purpose and Need for an Authorization, and the Reasonable Alternatives Considered.

CEQ proposed to revise section 1502.13 of the implementing regulations, which is the purpose and need section of an EIS that sets forth the rationale for the agency's proposed action. *Id.* at 55,760. The current regulations specifically direct agencies to base the purpose and need assessment on the goals of the applicant and an agency's statutory authority. Also, the current regulations contain conforming changes were made to section 1508.1(z), the definition of reasonable alternatives. *Id.* CEQ is concerned that the current definition requires an agency to prioritize the need of an applicant over all other relevant factors and does not believe this is the best reading of NEPA statutory language. Thus, CEQ is proposing to revert to the language contained in the 1978 regulations, which allowed agencies to consider a variety of factors. *Id.*

As an initial matter, whether explicitly or implicitly stated, the objectives and goals of the applicant are relevant factors that agencies should consider as part of a holistic environmental review process.

More critical, however, is how the proposed changes to the purpose and need assessment affect the set of reasonable alternatives an agency may consider. As part of an efficient environmental review process, CEQ should consider limiting the scope of reasonable alternatives that agencies consider by having agencies analyze a reasonable—but not exhaustive—number of alternatives as part of the NEPA process. Such a limitation is reasonable and consistent with NEPA's requirements because it directs agencies to approach NEPA analyses in an efficient manner by analyzing alternatives most likely to be relevant alternatives, not unlikely or unreasonable

potential choices that could be inconsistent with the proponent's goal and objectives. The experience of most EEI members is that most proposed projects have multiple alternatives available to them in terms of route taken and method of crossing utilized. However, these are not available in every instance and often these projects are limited in terms of truly reasonable alternatives—e.g., one route going over more public lands, another going over more private lands, or a route utilizing fewer stream crossings than a different route. Analyzing a reasonable number of alternatives—based on the applicant's purposes and need for a project—can avoid broad and inefficient analyses. NEPA analyses should allow agencies to efficiently and effectively permit infrastructure projects that will enable the industry to continue its clean energy transformation.

IV. CEQ Should Strive for Consistency Among Agency NEPA Regulations.

CEQ proposes to revise section 1507.3(a) and (b) to clarify that while agency specific NEPA regulations must be consistent with CEQ's regulations, agencies have the discretion and flexibility to develop procedures beyond the CEQ regulatory requirements, enabling agencies to address their specific programs and the contexts in which they operate. *Id.* at 55,761.

As discussed *supra*, EEI members are in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation includes the construction and maintenance of critical infrastructure projects, which frequently require federal permits from multiple federal agencies. In order to facilitate this clean energy transformation, as agencies issue their own NEPA regulations, providing consistency would enable a more efficient permitting process. Also, where multiple federal agencies are involved, ensuring that the NEPA regulations of the lead agency are controlling would also increase permitting efficiency and allow project

applicants additional certainty and speed the permitting process. Providing consistency across agencies as they work on permitting projects in the NEPA process that will help drive the industry's clean energy transformation will allow for these projects to move forward efficiently.

V. The Challenges of GHG Emissions Quantification Remain; Agencies Undertaking NEPA Analyses Should Do Robust Analyses and Place Those Analyses in Context.

CEQ proposes to revise the current regulatory definition of “effects.” The current definition removed the historic and explicit distinction between direct and indirect effects, eliminating the requirement to address “cumulative effects,” instead requiring analysis of only those effects that are “reasonably foreseeable” and have a “reasonably close causal relationship” with the agency’s proposed action. CEQ proposes to revert to the 1978 regulatory language that requires analysis of direct, indirect, and cumulative effects of an action. *Id.* at 55, 762.

EEI provided comments in 2010, 2015, and 2019, and 2020 on the CEQ’s previous proposals addressing GHG emissions in NEPA analyses. In these comment letters, EEI encouraged CEQ to balance the information disclosure requirements of NEPA with the inherent difficulties of quantifying reasonably foreseeable direct and indirect GHG emissions related to a specific proposed action or project. Of particular concern to EEI members was the risk that agencies would engage in speculative emissions analyses, even when potential emissions might not be significant.⁸

GHG analyses remain important and must be placed in context. NEPA requires federal agencies to consider environmental impacts of proposed actions as part of agencies’ decision-making

⁸EEI’s previous comments are attached as Appendix A.

processes. NEPA is not, in and of itself, determinative of project approval as it simply prescribes the steps by which agencies must evaluate options and explain their decision-making. Thus, while GHG emissions alone cannot and should not drive federal NEPA decision-making, agencies should take steps to reasonably quantify GHG emissions—and appropriately contextualize and caveat such quantifications—to satisfy NEPA’s informational requirements and to minimize challenges to environmental reviews on the grounds that they did not adequately address such emissions.

A. Considering GHG Emissions is Appropriate and Often Required by NEPA.

GHG emissions are relevant to NEPA analyses, and considering such emissions is appropriate in the context of environmental reviews designed to ensure well-informed decision-making. In prior comments, EEI encouraged CEQ to balance the information disclosure requirements of NEPA with the inherent difficulties of quantifying reasonably foreseeable direct and indirect GHG emissions related to a specific proposed action or project.

CEQ should consider, where appropriate, indirect or cumulative effects—including GHG emissions—as these effects may be relevant to selection of an alternative and courts have found failure to consider them a violation of NEPA. Accordingly, a reasonable and legally defensible reading of the statute requires that, where appropriate, NEPA analyses include a consideration of the indirect and cumulative impacts of GHG emissions from a project. Moreover, other environmental, contextual, and statutory factors may dictate that more detailed GHG emissions analyses be undertaken, particularly if they help to put emissions in the appropriate context and may help projects withstand judicial delays. This is a key consideration for electric companies

since legally insufficient NEPA analyses can lead to costly delays in the permitting of critical energy infrastructure and cause investor uncertainty.

B. Sector-Wide Analysis of GHG Emissions Can Help Support the Continued Transformation of the Power Sector.

CEQ should encourage agencies to consider existing emissions estimates where relevant, including sector-wide estimates, when assessing GHG emissions impacts. Sector-wide contextualization is particularly important when agencies must undertake NEPA GHG analyses for specific energy projects.

GHGs, unlike other pollutants, are a well-mixed pollutant, meaning that the amount measured in the atmosphere is roughly the same all over the world, regardless of the location of any particular source of emissions. Given the well-mixed nature of GHGs, agencies should have the ability to consider power sector reductions in both GHG and other emissions for projects or proposed actions that implicate electricity generation, transmission, or distribution when reasonable and suggested by the project proponent. Sector-wide assessments of total GHG emissions and emissions trends can help federal agencies and interested stakeholders better understand the potential direct, indirect, and cumulative emissions impacts of a proposed action. For example, sector-wide assessments can help demonstrate that a particular project—such as a new electric transmission line or natural gas pipeline—serves to reduce indirect emissions by allowing electric companies to integrate more renewable resources or shift fuel sources.

Where GHG emissions are difficult to quantify, sector-wide emissions assessments can provide helpful context that allows agencies to provide a qualitative assessment of the potential impacts of a proposed permit or action. Reliable sector-wide assessments can help inform both agency

decision-making and interested stakeholders consistent with the requirements and objectives of NEPA. CEQ should encourage agencies to adopt this approach when reasonable.

VI. Conclusion.

As discussed above, EEI members have a compelling interest in the revisions to the CEQ NEPA implementing regulations. An efficient NEPA review process, which includes tailored and focused considerations of reasonable alternatives, consistent NEPA regulations across multiple federal agencies, and appropriate consideration and contextualization of GHG emissions, will enable EEI members to continue the clean energy transformation. Questions on these comments may be directed to [Alex Bond](#) (202-508-5523), [Eric Holdsworth](#) (202-508-5103) or [Riaz Mohammed](#) (202-508-5036).

Appendix A



Edison Electric
INSTITUTE

Quinlan J. Shea, III

Vice President, Environment, Natural Resources,
and Occupational Safety & Health

March 10, 2020

Mr. Edward A. Boling
Ms. Viktoria Z. Seale
Council on Environmental Quality
730 Jackson Place NW
Washington, DC 20503

[Submitted Electronically]

RE: Comments of the Edison Electric Institute on the Council on Environmental Quality's Update to the Regulations Implementing the Procedural Provisions of the National Environmental Policy Act

Dear Mr. Boling & Ms. Seale:

The Edison Electric Institute (EEI) appreciates the opportunity to submit comments on the advanced notice of proposed *Update to the Regulations Implementing the Procedural Provisions of the National Environmental Policy Act* issued by the Council on Environmental Quality (CEQ). 85 *Fed. Reg.* 1,684 (Jan. 10, 2020) (Proposed NEPA Rule). The Proposed NEPA Rule is intended to make the regulations more efficient and effective, so that federal agencies conducting NEPA reviews may conduct them in a timelier and less resource intensive manner, while still ensuring that agencies consider the wide range of environmental impacts of proposed actions.

EEI is the association that represents all U.S. investor-owned electric companies. Our members provide electricity for more than 220 million Americans and operate in all 50 states and the District of Columbia. As a whole, the electric power industry supports more than seven million jobs in communities across the United States. The electric power industry is in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation is being driven by a wide range of factors, including: low and declining costs for natural gas and renewable energy resources, technological improvements, changing customer expectations, federal and state regulations and policies, and the increasing use of distributed energy resources (DERs). As a result, the mix of resources used to generate electricity has changed dramatically over the last decade and is increasingly clean.

Concurrent with this transition, EEI member companies are investing significant amounts of capital—estimated at more than 130 billion dollars in 2019 alone—to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure in order to integrate

202-508-5027 | qshea@eei.org

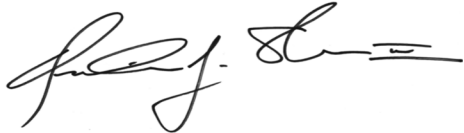
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and deliver a balanced mix of resources from both centralized generation sources and DERs to customers.

EEI's comments focus on three primary areas: first, the industry's continued clean energy transformation and the resultant greenhouse gas (GHG) emission reductions; second, several of the process-oriented changes in the Proposed NEPA Rule that would make the NEPA review process more efficient, effective and timely, while ensuring that NEPA's larger goals of informing the public and promoting federal decisions that consider environmental impacts are met; and, third, that agencies should take steps to quantify GHG emissions reasonably—and appropriately contextualize and caveat such quantifications—to satisfy NEPA's informational requirements as well as minimize challenges to environmental reviews on the grounds that they did not adequately address such emissions.

EEI appreciates the opportunity to comment on the Proposed NEPA Rule. Questions may be directed to Alex Bond (abond@eei.org, 202-508-5523), Eric Holdsworth (eholdsworth@eei.org, 202-508-5103) or Sarah Ball (sball@eei.org, 202-508-5208).

Sincerely,

A handwritten signature in black ink, appearing to read "Quinlan J. Shea, III". The signature is fluid and cursive, with a long horizontal line extending from the end.

Quinlan J. Shea, III

**COMMENTS OF THE EDISON ELECTRIC INSTITUTE
UPDATE TO THE REGULATIONS IMPLEMENTING THE PROCEDURAL
PROVISIONS OF THE NATIONAL ENVIRONMENTAL POLICY ACT**

DOCKET NO. CEQ-2019-0003

March 10, 2020

The Edison Electric Institute (EEI) appreciates this opportunity to submit comments on the proposed rule *Update to the Regulations Implementing the Procedural Provisions of the National Environmental Policy Act* issued by the Council on Environmental Quality (CEQ). 85 *Fed. Reg.* 1,684 (Jan. 10, 2020) (Proposed Rule or Proposed NEPA Rule). The Proposed NEPA Rule is intended to make the regulations more efficient and effective, so that federal agencies conducting NEPA reviews may conduct them in a timelier and less resource intensive manner and ensure that that agencies consider the wide range of environmental impacts of proposed actions. *See id.* at 1,685.

EEI is the association that represents all U.S. investor-owned electric companies. Our members provide electricity for more than 220 million Americans and operate in all 50 states and the District of Columbia. As a whole, the electric power industry supports more than seven million jobs in communities across the United States. EEI member companies invest more than \$110 billion annually to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure in order to provide affordable and reliable electricity to customers.¹ Such critical infrastructure projects frequently require federal permits and are therefore subject to NEPA

¹ *See* EEI, Industry Data, Statistical Highlights: Capacity and Generation (2018), <http://www.eei.org/resourcesandmedia/industrydataanalysis/industrydata/Pages/default.aspx>.

reviews, which increasingly have become the focus of protracted litigation. Litigation over NEPA reviews significantly adds to the time and costs of critical infrastructure projects. As the proponents of such projects, EEI's member companies have a compelling interest in the Proposed NEPA Rule's efforts to assist agencies in the efficient, timely issuance of environmental reviews while ensuring that agencies take a "hard look" at environmental consequences as mandated by NEPA.

EEI's comments focus on three primary areas: first, the industry's continued clean energy transformation and the resultant greenhouse gas (GHG) emission reductions; second, several of the process-oriented changes in the Proposed NEPA Rule that would make the NEPA review process more efficient, effective and timely, while ensuring that NEPA's larger goals of informing the public and promoting federal decisions that consider environmental impacts are met; and, third, that agencies should take steps to quantify GHG emissions reasonably—and appropriately contextualize and caveat such quantifications—to satisfy NEPA's informational requirements as well as minimize challenges to environmental reviews on the grounds that they did not adequately address such emissions.

I. Electric Companies Continue To Lead the Clean Energy Transition And Reduce GHG Emissions.

The electric power industry is in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation is being driven by a wide range of factors, including: declining costs for natural gas and renewable energy resources; technological improvements; changing customer expectations; federal and state regulations and policies; and, the increasing use of distributed energy resources. As a result, the mix of resources used to generate electricity in the United States has changed dramatically over the last decade

and is increasingly clean. In 2018, natural gas surpassed coal as the main source of electricity generation in the United States for the third year in a row, with natural gas-based generation powering 35.2 percent of the country's electricity and non-emitting generation comprising 37.2 percent of all generation—compared to coal-based generation at 27.5 percent.² Since 2014, more than half of the industry's investments in new electricity generation have been in wind and solar generation resources,³ and nearly 40 percent of America's electricity now comes from carbon-free resources, including nuclear power, hydropower, solar, and wind.⁴ Further, the Energy Information Administration (EIA) conservatively projects that the United States will add 72 gigawatts (GW) of new wind and solar capacity between 2018 and 2021 alone and that, long-term, demand for new electric generating capacity will be met by renewables and efficient natural gas as older coal-based and less-efficient natural gas-based generating units retire.⁵ States are expected to rely on the electric power sector for electrification of other sectors to achieve clean energy and climate goals, as mitigating the impacts of climate change continues to be an essential public policy question.

² See Department of Energy, EIA, *What is U.S. Electricity Generation By Energy Source* (Oct. 2019), <https://www.eia.gov/tools/faqs/faq.php?id=427&t=3>.

³ See EIA, *Nearly Half of Utility-Scale Capacity Installed in 2017 Came from Renewables* (Jan. 10, 2018), <https://www.eia.gov/todayinenergy/detail.php?id=34472>. See also EEI, *Industry Data, Statistical Highlights: Capacity and Generation* (2018), <http://www.eei.org/resourcesandmedia/industrydataanalysis/industrydata/Pages/default.aspx>.

⁴ See EIA, *Electricity Explained: Electricity in the United States* (Apr. 2018), https://www.eia.gov/energyexplained/index.php?page=electricity_in_the_united_states.

⁵ See EIA, *Annual Energy Outlook 2019: With Projections to 2050* (Jan. 24, 2019) at 94, 96, <https://www.eia.gov/outlooks/aeo/pdf/aeo2019.pdf>. While EIA notes in the *Annual Energy Outlook* (AEO) that the amount of renewable and natural gas-based generation deployed are dependent on the price of natural gas, this does not impact the expected closure of coal-based and other less efficient generation. See *id.* at 108.

These changes have had a beneficial impact on the sector's GHG emissions. As of the end of 2018, the electric power sector has reduced its carbon dioxide (CO₂) emissions by 27 percent compared with near peak levels in 2005—the lowest level in nearly 30 years.⁶ EEI's member companies have reduced their CO₂ emissions even more dramatically to approximately 37 percent below 2005 levels.⁷ And these reductions will continue: EEI's member companies are on a path to reduce CO₂ emissions nearly 50 percent by 2030 and potentially 80 percent or more by 2050 compared to 2005 levels while keeping electricity affordable and reliable.⁸

Other environmental benefits of this clean energy transition include significant reductions in criteria pollutants, acid gases and other hazardous air pollutants (HAPs), like mercury. As of 2018, emissions of sulfur dioxide (SO₂) and nitrogen oxides (NO_x) have been reduced by 92 and 84 percent, respectively, from 1990 levels.⁹ The industry also has reduced mercury emissions by approximately 86 percent from 2010 to 2017, and emissions of both acid gases and of total HAPs were reduced by 96 percent.¹⁰ This also has led to significant improvements in the sector's

⁶ Final data for 2019 is not yet available but is expected to show continued decreases in CO₂ emissions from electricity generation.

⁷ Based on an EEI analysis of CO₂ emissions data from ABB Velocity Suite.

⁸ Many EEI member companies have announced significant voluntary commitments to further reduce CO₂ emissions between 2030 and 2050, many of which aim to reduce emissions 80 percent below 2005 levels by 2050. These commitments are attached here as Appendix A.

⁹ See U.S. Environmental Protection Agency (EPA), Power Plant Emission Trends, <https://www.epa.gov/airmarkets/power-plant-emission-trends>.

¹⁰ See EPA, *National Emission Standards for Hazardous Air Pollutants: Coal- and Oil-Fired Electric Utility Steam Generating Units—Reconsideration of Supplemental Finding and Residual Risk and Technology Review*, 84 *Fed. Reg.* 2,670, 2,689 (Feb. 7, 2019).

impact on water resources as the volume of chemical discharges by the electric sector were nearly 82 percent less in 2018 as compared to 2005 levels.¹¹

II. The Proposed NEPA Rule Makes Several Well-Supported And Necessary Process Changes That Could Facilitate More Efficient, Effective, And Timely NEPA Reviews.

NEPA establishes procedural requirements for proposed major federal actions, including preparation of a statement on: (i) the environmental impact of the proposed action, (ii) any adverse environmental effects which cannot be avoided should the proposal be implemented, (iii) alternatives to the proposed action, (iv) the relationship between local short-term uses of man's environment and the maintenance and enhancement of long-term productivity, and (v) any irreversible and irretrievable commitments of resources which would be involved in the proposed action should it be implemented.¹² As a result of historical CEQ regulations and guidance—as well as Supreme Court and other court decisions—NEPA environmental reviews have become characterized by longer, more resource intensive process and significant delays, especially for proponents building critical infrastructure projects. Overall, improvements to the NEPA regulations that could facilitate more efficient, effective and timely NEPA reviews by federal agencies are necessary and appropriate and CEQ should codify these practices in the final regulations. The following highlights several specific revisions to the NEPA regulations that could make the NEPA review process more efficient, effective and timely, while ensuring that NEPA's larger goals of informing the public and promoting federal decisions that consider environmental impacts are met.

¹¹ Based on an EEI analysis of overall discharges reported in EPA's Toxic Release Inventory (TRI) program for 2005 and 2018 from coal- and oil-based generating facilities.

¹² See 42 U.S.C. § 4332(2)(C).

A. CEQ Appropriately Proposes to Provide Federal Agencies With the Flexibility to Allow Project Proponents to Assist in the Preparation of Environmental Reviews.

CEQ proposes to amend their regulations to allow project proponents and contractors to assume a greater role in preparing environmental documents that would form the basis for the environmental review. *See* 85 *Fed. Reg.* at 1,699. CEQ proposes new section 1501.9 on “scoping” in order to allow agencies to begin the scoping process as soon as any proposed action is “sufficiently developed for meaningful agency consideration.” *Id.* CEQ’s proposal would allow project proponents to participate in the scoping process in advance of the agency’s notice of intent (NOI) to prepare an environmental impact statement (EIS), which under the current regulations is the first time a project proponent could comment on an agency’s scoping decision. *Id.* It is worth noting that allowing for increased participation in the scoping process would not displace the need for robust evaluation of project alternatives and potential impacts.

The ability of project proponents and consultants to participate in the preparation of materials as part of the environmental review is a long-standing and well-established practice that benefits the federal agencies, the project proponents, and the public. The Proposed NEPA Rule formalizes this by allowing project proponents to participate in the preparation of both Environmental Assessments (EAs) and EIS’s as well in proposed section 1506.5(c). EEI members and their contractors not only are in the best position to provide critical information about a proposed projects, alternatives, and potential effects, but also possess significant environmental, technical, resources and engineering information that could enable more efficient and timely preparation and evaluation of environmental reviews. Allowing project proponents to participate earlier in

the scoping and pre-scoping process—especially before the publication of a NOI—will provide the agencies with a more robust record and understanding of the scope and needs of each project.

As a result of this early participation, all stakeholders will be able to evaluate a wider array of data on projects and provide more robust public comments. Absent proposed section 1501.9 and 1506.5(c), there is no formal way for project proponents to participate in the pre-scoping process until publication of a NOI. This will necessarily result in improved communication between the project proponent and the federal agency, which will improve the quality of NEPA documents as well as the efficiency of the NEPA process. Importantly, while project proponents and consultants would have a greater role in preparation in the documents that form the basis for the environmental review, the federal agencies would maintain the responsibility to independently evaluate and take responsibility for the documents. *See id.* It is worth noting that the resulting EIS, EA, or Finding of No Significant Impacts (FONSI) still will be subject to public comment, such that these decisions will be subject to rigorous review by all stakeholders—including the public. CEQ should finalize this codification in the final regulations.

In addition to amending the regulations to allow project proponents to participate earlier in the NEPA process, CEQ should give consideration to expanding these changes to proactively include the ability for the project proponent to coordinate with the NEPA contractors selected by the federal agency in charge of the NEPA review process. Often, the contractor is barred from coordinating with the non-federal applicant before a permit application has been submitted to the lead agency. As discussed *supra*, EEI members and their contractors are in the best position to provide timely, accurate, and relevant information regarding a proposed project and its effects.

The prohibition on coordinating with contractors selected by the lead agency can therefore result in a significant lag in completing draft NEPA documents, an increase in the cost of preparation, and a greater likelihood that the preliminary draft document will contain inaccuracies that could have otherwise been cured through coordination with the project proponent. CEQ should consider ways to encourage coordination between contractors and project proponents.

B. CEQ Appropriately Recognizes the Importance of Categorical Exclusions and Provides Needed Clarity on Their Use; Adopting CEs Across Agencies Will Serve the Purposes of NEPA and Promote Efficiency.

Agencies can categorically exclude actions from a detailed review if the action normally would not have significant effects, meaning that they would not require an EA (or, ultimately, an EIS) to satisfy NEPA's requirements that significant impacts be considered during the federal decision-making process.¹³ These exclusions must be informed by their own NEPA analyses, are subject to public notice and comment, and are reviewed by CEQ, meaning that they are vetted by all stakeholders. *See* 40 C.F.R. § 1507.3(a). Once established, categorical exclusions (CEs) commonly are used by federal agencies to satisfy expeditiously NEPA requirements for certain types of actions, but CEQ has only recently provided guidance on the proper development and implementation of CEs and this guidance has not been incorporated into the NEPA regulations. EEI's members consistently rely on CEs to authorize a variety of routine operation and maintenance activities related to the electric power industry.

¹³ *See* 40 C.F.R. § 1508.4. Agencies can comply with the procedural requirements of NEPA through three primary environmental documents: a CE, an EA or an EIS. CEs are the most efficient form of NEPA compliance for project proponents since they establish early in the process that a proposed action will not have a significant impact on the environment. Since CEs do not require further analysis when utilized by a project proponent, CEs are subject to a robust process upon their adoption, which includes consultations between agencies and CEQ, as well as public comment on the proposed CE.

To address this and other concerns, CEQ proposes to update its regulations to clarify the use of CEs, proposing an entirely new section of regulations that will outline the process by which an agency considers whether a CE applies to a proposed action. *See* 85 *Fed. Reg.* at 1,695; *see also* proposed 40 C.F.R. 1501.4. Importantly, the proposed regulation would provide additional guidance on the consideration of extraordinary circumstances and clarify that the presence of an extraordinary circumstance does not preclude the use of a CE, but that the federal agency should consider whether mitigating circumstances justify categorically excluding the proposed project. *See* 85 *Fed. Reg.* at 1,695.

CEQ also proposes two avenues for expanding the use of CEs. First, proposed section 1506.3(f) explicitly addresses an agency's authority to adopt another's determination that a CE applies to a specific project if the adopting agency's proposed action is substantially the same. *See id.* at 1,706. Second, proposed section 1507.3(e)(5) would go further, allowing agencies to import wholesale a CE established under another agency's NEPA procedures into its own NEPA procedures and regulations. *See id.* at 1,707. Both of these proposals should be finalized.

It is worth noting that, as agencies develop their own CEs, each undergoes a comprehensive review of the potential impacts of an action of the type to be excluded and determines that there will be no significant impact. CEQ's first proposed option, allowing for the utilization of a CE for a particular action that has already been reviewed and determined to have no significant effect by another agency will make the NEPA process more efficient and will save taxpayer resources. To make this authority clear in any final regulations, CEQ should consider including a cross reference to section 1506.3(f) in section 1507.3. There is no added benefit to having an

agency complete an EA or EIS for a proposed project or action if another federal agency already has determined that it should be categorically excluded from such reviews because it will not have, individually or cumulatively, a significant effect on the human environment.¹⁴

Proposed section 1507.3 will likely introduce considerable efficiencies by allowing different agencies to formally adopt CEs determined as appropriate by other expert agencies into their own NEPA regulations. For example, such adoption may be particularly useful for the various entities under the Department of the Interior (DOI), which should not have to duplicate work already undertaken by other DOI entities. These proposed revisions to utilizing CEs make sense and are consistent with the goals of making NEPA reviews more efficient and effective. CEQ should finalize this provision and encourage agencies to adopt CEs from other agencies into their own NEPA regulations.

C. Utilizing the Tiering Process and Allowing for Incorporation by Reference Could Make the NEPA Review Process More Efficient.

CEQ also proposes revisions to clarify when agencies can use existing studies and environmental analyses in the NEPA process—called “tiering.” *See id.* at 1,700. Federal agencies utilizing the tiering process to build upon previously conducted environmental studies and analyses, as well as earlier decisions made during the federal, state, or local public reviews, is reasonable and appropriate for improving the efficiency of the overall review process. Likewise, federal agencies should incorporate by reference relevant earlier studies and analyses that are relevant to the current proposed project, thereby building upon previous work in an efficient manner without

¹⁴ CEs are not exemptions from NEPA and CEQ’s guidance on CE requires that such exclusions be well substantiated, consistent with reasoned agency decision-making. *See CEQ, Final Guidance on Federal Departments and Agencies on Establishing, Applying, and Revisions Categorical Exclusions under the National Environmental Policy Act*, 75 Fed. Reg. 75,628.

requiring new processes or materials that do not substantively add any new information to the decision-making and review process.

Requiring federal agencies to tier and incorporate by reference would minimize duplication of analysis on subjects where another agency may be the subject matter expert. This will ensure that the lead agency is provided, and is utilizing, the most appropriate information in making their decision. Tiering and incorporating by reference are proven and effective methods of making the NEPA review process more efficient, and CEQ should finalize the proposed changes to sections 1501.11 and 1501.12 in the final regulations. *Id.* at 1,695.

CEQ also should consider whether to provide additional guidance on how to appropriately tier off of a previous study as well as whether to create a “library” or repository of earlier studies that is available to the federal agencies as well as interested parties in order to allow project proponents and agencies to quickly and efficiently find relevant material that can be utilized during the review process.

D. Clarifications Regarding Alternatives.

The Proposed NEPA Rule contains proposed changes to section 1502.14 regarding the clarity and scope of what project alternatives agencies should consider as part of their NEPA reviews. CEQ notes that NEPA does not specifically mandate which set of “alternatives” must be considered when analyzing a proposal, and that NEPA section 102(2)(C) provides only that an agency should prepare a detailed statement addressing “alternatives to the proposed action” *Id.* at

1,702, citing 42 U.S.C. 4332(2)(C).¹⁵ The Proposed NEPA Rule notes that agencies still must consider alternatives in their analyses of projects, and that those analyses must be reasonable and that the reasonableness of these analyses are resolved “not by any particular number of alternatives considered, but by the nature of the underlying agency action.” CEQ further notes that the analysis of alternatives need not be exhaustive but must provide information “sufficient to permit a reasoned choice of alternatives for the agency to evaluate reasonable available alternatives.” *Id.* Importantly, the Proposed NEPA Rule clarifies that when an agency is responsible for reviewing applications for authorization, agency’s should base the purpose and need statement—and therefore the resulting alternatives—on both the applicant’s goals for the projects and the underlying statutory authority for the agency.

CEQ’s proposal to have agencies analyze a reasonable but not exhaustive number of alternatives as part of the NEPA process is eminently reasonable and consistent with NEPA’s requirements. Analyzing an exhaustive number of potential alternatives is not an efficient use of government resources, as most projects have multiple alternatives available to them in terms of route taken and method of crossing utilized. However, these are not unique in every instance and often these projects are limited in terms of truly reasonable alternatives—e.g., one route going over more public lands, another going over more private lands, or a route utilizing fewer stream crossings than a different route. Analyzing a reasonable number of alternatives based on the applicant’s purposes and need for a project to avoid exhaustive analysis to more efficiently utilize government resources is a laudable goal. CEQ should finalize the proposed changes. CEQ should

¹⁵ NEPA Section 102(2)(E) requires only that agencies “study, develop, and describe appropriate alternatives to recommended courses of action.” 42 U.S.C. 4332(2)(E).

not, however, establish a maximum number of alternatives that may be considered or otherwise constrain the consideration of actual and feasible alternatives.

E. CEQ Should Formally Adopt And Build On The One Federal Decision And Joint Decision Documents.

CEQ also proposes to adopt and build on the “One Federal Decision” framework by requiring agencies to prepare joint EISs/Records of Decision (RODs) and EAs/FONSIs when practicable. *See id.* at 1,691. One Federal Decision is an effective framework to help agency cooperation, and formally requiring joint preparation of EISs and RODs—as well as EAs and FONSI—would effectively and efficiently require agencies to produce joint decisions that can efficiently and robustly make use of the resources of the expert agencies, as proposed section 1500.6 would do. One Federal Decision and the proposed modifications in the Proposed NEPA Rule are well founded and CEQ should finalize them.

CEQ also proposes to codify in section 1501.7 the practice of designating lead and cooperating agencies in EAs, and to require agencies to cooperate, to the fullest extent possible, with state, tribal, and local governments. *See id.* at 1,698. While the joint decision document framework remains relatively new, it has been successfully implemented in a small number of RODs.¹⁶ These proposed changes should promote better and earlier coordination among various levels of government agencies and should reduce duplicative, inconsistent reviews and decisions, and, ultimately, provide for faster project implementation. CEQ should finalize these proposed changes as well.

¹⁶ This framework has never been applied to FONSI, which could lead to initial agency coordination issues that are likely to lessen with time and ongoing implementation of these changes.

The Proposed NEPA Rule also would revise section 1506.2 to acknowledge the increasing number of state, tribal and local governments preparing environmental documents, including those prepared pursuant to NEPA and avoid duplication of effort. Where state laws, tribal laws, or local ordinances have environmental impact statement or similar requirements in addition to but not in conflict with those in NEPA, federal agencies may cooperate in fulfilling these requirements, as well as those of federal laws, so that one document will comply with all applicable laws. *Id.* at 1,704. This proposed change would help increase efficiency in the NEPA process in states that have existing and robust state level environmental documentation requirements that have significant overlap with NEPA resulting in inefficient and inconsistent processes and outcomes.¹⁷ CEQ should finalize the proposed changes and consider further if adoption of state level environmental documents by a federal agency under section 1506.3 would be feasible.

III. The Challenges of GHG Emissions Quantification Remain; Agencies Undertaking NEPA Analyses Should Do Robust Analyses And Place Those Analyses In Context.

EEI provided comments in 2010, 2015, and 2019 on the agency's previous proposals addressing GHG emissions in NEPA analyses.¹⁸ In these comments, EEI encouraged CEQ to balance the information disclosure requirements of NEPA with the inherent difficulties of quantifying reasonably foreseeable direct and indirect GHG emissions related to a specific proposed action or project. Of particular concern to EEI members was the risk that agencies would engage in

¹⁷ See, e.g., the California Environmental Quality Act, California Public Resources Code, Sections 21000 - 21178, and Title 14 CCR, Section 753, and Chapter 3, Sections 15000 – 15387.

¹⁸ EEI's previous comments are attached here as Appendices B, C, and D.

speculative emissions analyses, even when potential emissions might not be significant. While CEQ states that it is not addressing GHG analyses separately in these proposed regulations, see 85 Fed. Reg. at 1,710, many of the issues raised in the Proposed Rule will have implications for NEPA reviews that address GHG emissions. Accordingly, these comments address these implications with the goal of balancing reasonable emissions assessments with judicial certainty for proponents of critical energy infrastructure.

GHG analyses remain important and must be placed in context. NEPA requires federal agencies to consider environmental impacts of proposed actions as part of agencies' decision-making processes. NEPA is not, in and of itself, determinative of project approval as it simply prescribes the steps by which agencies must evaluate options and explain their decision-making. Thus, while GHG emissions alone cannot and should not drive federal NEPA decision-making, agencies should take steps to reasonably quantify GHG emissions—and appropriately contextualize and caveat such quantifications—to satisfy NEPA's informational requirements and to minimize challenges to environmental reviews on the grounds that they did not adequately address such emissions.

A. Considering GHG Emissions is Appropriate and Often Required by NEPA.

GHG emissions are relevant to NEPA analyses, and considering such emissions is appropriate in the context of environmental reviews designed to ensure well-informed decision-making. In prior comments, EEI encouraged CEQ to balance the information disclosure requirements of NEPA with the inherent difficulties of quantifying reasonably foreseeable direct and indirect GHG emissions related to a specific proposed action or project. As the Proposed Rule notes, many NEPA reviews become mired in efforts to define what would constitute a reasonably foreseeable

direct, indirect, and cumulative impact of a proposed action. *See id.* at 1,708. To that end, CEQ’s proposal to clarify the definition of “effects” such that the emphasis is less on categorizing effects is appropriate. *See id.* at 1,707-08. Requiring agencies to apply a “reasonableness” standard for determining the scope of those reviews also is an appropriate approach. However, it may not be reasonable or defensible to entirely exclude indirect or cumulative effects—including GHG emissions—from consideration. As CEQ noted in its *Draft National Environmental Policy Act Guidance on Consideration of Greenhouse Gas Emissions*, GHG emissions are “inherently a cumulative effect.” 84 *Fed. Reg.* 30,098 (June 26, 2019). These effects may be relevant to selection of an alternative and courts have found failure to consider them a violation of NEPA.¹⁹ Accordingly, a reasonable and legally defensible reading of the statute requires that, where appropriate, NEPA analyses include a consideration of the indirect and cumulative impacts of GHG emissions from a project.

Reasonableness, at times, may still require an assessment of effects that would have been categorized in the past as “indirect” or “cumulative.” Moreover, other environmental, contextual, and statutory factors may dictate that more detailed GHG emissions analyses be undertaken. CEQ’s proposed regulations, therefore, should not be read as a bar to such analyses, particularly if they help to put emissions in the appropriate context and may help projects withstand judicial delays because no GHG emissions analyses were conducted. This is a key consideration for electric companies since legally insufficient NEPA analyses can lead to costly delays in the permitting of critical energy infrastructure and cause investor uncertainty. CEQ’s final rule or

¹⁹ *See e.g., WildEarth Guardians v. BLM*, 870 F.3d 1222 (10th Cir. 2017); *High Country Conservation Advocates v. USFS*, 52 F. Supp. 3d 1174 (D. Col. 2014).

subsequent guidance should help agencies navigate the various judicial decisions finding that failure to quantify GHG emissions from proposed actions and projects violates NEPA.

Recognizing that GHG emissions alone should not drive federal NEPA decision making, it is critical to EEI's members that CEQ clarify in which situations appropriate agencies should take reasonable steps to quantify GHG emissions—and appropriately contextualize and caveat such quantifications—to satisfy NEPA's aim of providing information and soliciting public input.

Even if the proposed clarification of “effects,” is finalized, many prior court decisions address the extent to which, and the methods through which, agencies may need to consider the indirect and cumulative impacts associated with reasonably foreseeable direct or indirect GHG emissions. Legal decisions make clear that the quantification of such emissions and an assessment of their potential impacts is a very fact-specific endeavor, which turns not only on the proposed action, but also on the scope of the agency's authority and the information that it has available to it at the time of the analysis.²⁰

Accordingly, if the goal is legally sufficient environmental analyses, it may be helpful for CEQ to provide agencies that may be required to engage in such analyses with general guidance on how to approach indirect and cumulative impacts for GHG emissions in the development of NEPA analyses. Where such reviews may be helpful or required, it is appropriate that CEQ

²⁰ See, e.g., *WildEarth Guardians v. Zinke*, No. 1:16-cv-1724 (D.D. C. Mar. 19, 2019) (finding that the Bureau of Land Management (BLM) was required to consider in its NEPA analysis the reasonably foreseeable indirect emissions from oil and gas leasing but remanding to BLM to determine whether and how to attempt quantification of those emissions).

encourage agencies to consider existing emissions estimates, including sector-wide estimates, when assessing indirect and cumulative GHG emissions impacts.²¹

B. Sector-Wide Analysis of GHG Emissions Can Help Support the Continued Transformation of the Power Sector.

CEQ should encourage agencies to consider existing emissions estimates when relevant, including sector-wide estimates, when assessing GHG emissions impacts. Sector-wide contextualization is particularly important when agencies must undertake NEPA GHG analyses for specific energy projects. If not included in the final regulations, CEQ's guidance should note that the use of such analyses can be helpful and should be available to project proponents.

GHGs, unlike other pollutants, are a well-mixed pollutant, meaning that the amount measured in the atmosphere is roughly the same all over the world, regardless of the location of any particular source of emissions. Given the well-mixed nature of GHGs, agencies should have the ability to consider power sector reductions in both GHG and other emissions for projects or proposed actions that implicate electricity generation, transmission, or distribution when reasonable and suggested by the project proponent.

Sector-wide assessments of total GHG emissions and emissions trends can help federal agencies and interested stakeholders better understand the potential direct, indirect, and cumulative emissions impacts of a proposed action. For example, sector-wide assessments can help demonstrate that a particular project—such as a new electric transmission line or natural gas pipeline—actually serves to reduce indirect emissions by allowing electric companies to

²¹ To this end, it also may be helpful for CEQ to provide examples of how agencies could address whether emissions are significant for NEPA purposes.

integrate more renewable resources or shift from higher-emitting coal to lower-emitting natural gas as a fuel.²²

Where GHG emissions are difficult to quantify, sector-wide emissions assessments can provide helpful context that allows agencies to provide a qualitative assessment of the potential impacts of a proposed permit or action. The effects of GHG emissions from any particular project on future warming trends cannot be predicted precisely with current scientific tools. However, reliable sector-wide assessments can help inform both agency decision-making and interested stakeholders consistent with the requirements and objectives of NEPA. CEQ should consider making clear in its final regulations that these assessments are available for consideration by agencies and can be put forward by project proponents.

²² While there may be concerns about project- or action-specific efforts to quantify GHG emissions, EIA's careful and well-substantiated analyses about sector-wide emissions unquestionably should satisfy NEPA's objectives.

APPENDIX A

EEI Member Company GHG Reduction Goals

COMPANY	GOAL
AEP (September 2019)	70% below 2000 levels by 2030 80% below 2000 levels by 2050
AES (November 2018)	- Reduce carbon intensity 50% from 2016 levels by 2022 - Reduce carbon intensity 70% from 2016 levels by 2030
Allele (June 2017)	40% below 2005 levels by 2030
Alliant (August 2018)	40% below 2005 by 2030 80% below 2005 by 2050
Ameren (September 2017)	35% below 2005 levels by 2030 50% below 2005 levels by 2040 80% below 2005 levels by 2050
AVANGRID (2016)	- Reduce CO₂ intensity 25% below 2015 levels by 2020 100% carbon neutral by 2035
Avista (April 2019)	- Carbon-neutral electricity by 2027 100% clean electricity by 2045
CenterPoint (February 2020)	70% below 2005 levels by 2035 - Natural gas CO₂ emissions 20-30% below 2005 by 2040⁵
CMS Energy (February 2020)	Net-zero CO₂ emissions by 2050
Dominion ¹ (February 2020)	65% below 2010 by 2030 (methane) 80% below 2010 by 2040 (methane) - Net-zero CO₂ and methane emissions by 2050

COMPANY	GOAL
DTE Energy (September 2019)	30% below 2005 levels by the early 2020s - At least 50% below 2005 levels by 2030 80% below 2005 levels by 2040 - Net-zero carbon emissions by 2050*
Duke Energy (September 2019)	- At least 50% below 2005 levels by 2030 - Attain net-zero CO₂ emissions by 2050*
Energys (March 2019)	20% below 2000 levels through 2020 - Reduce emissions intensity 50% below 2000 levels by 2030
Energys (November 2019)	80% below 2005 levels by 2050
Eversource ³ (December 2019)	Carbon neutral by 2030
Exelon ² (2018)	Additional 15% below 2015 levels by 2022 (operations-driven emissions)
FirstEnergy	90% below 2005 levels by 2045
Green Mountain Power (April 2019)	100% carbon-free electricity by 2025 100% renewable energy by 2030
Hawaiian Electric (May 2019)	100% renewable energy by 2045
Idaho Power (March 2019)	Provide 100% clean energy by 2045
Madison Gas & Electric (May 2019)	40% below 2005 levels by 2030 - Net-zero carbon electricity by 2050*

Note: See reverse for more

EEI Member Company GHG Reduction Goals (2)

COMPANY	GOAL
Montana-Dakota Utilities (2017)	Reduce GHG intensity 45% below 2005 levels by 2030
National Grid (November 2019)	45% below 1990 levels by 2020 70% below 1990 levels by 2030 - Net zero emissions by 2050
NextEra (June 2019)	Reduce CO₂ emissions rate 67% below 2005 levels by 2025
NiSource (May 2017)	50% below 2005 levels by 2025 - Reduce CO₂ emissions >90% by 2028 (NIPSCO, 11/18)
Northwestern ⁴ (December 2019)	Reduce carbon intensity 90% below 2010 levels by 2045
OG&E (December 2018)	40% below 2005 levels by 2019 50% below 2005 levels by 2030
PG&E (July 2018)	- Reduce 1 million tons of GHGs from company operations through 2022 40% below 1990 levels by 2030 80% below 1990 levels by 2050
Pinnacle West (2017)	- Reduce CO₂ emissions rate to <600 lbs./MWh by 2032 100% carbon-free electricity by 2050 (APS, 01/20)
PNM Resources (April 2019)	40% below 2012 levels by 2018 60% below 2012 levels by 2030 100% carbon-free electricity by 2040
Portland General Electric (February 2018)	More than 80% below 2010 levels by 2050
PPL (January 2018)	70% below 2010 levels by 2050

COMPANY	GOAL
PSEG (July 2019)	80% below 2005 levels by 2046 - Net-zero carbon emissions by 2050*
Puget Sound Energy (April 2019)	- Coal free by 2025 - Carbon neutral electric system by 2030 100% clean electricity by 2045
Southern California Edison (November 2019)	40% below 1990 levels by 2030 - Carbon neutrality by 2045
Southern Company (April 2018)	50% below 2007 levels by 2030 - Low- to no-carbon operations by 2050
TVA	60% below 2005 levels by 2020 75% below 2005 levels by 2030
Vectren (2016)	~60% below 2005 levels by 2024
WEC Energy Group (August 2018)	40% below 2005 levels by 2030 80% below 2005 by 2050
Xcel Energy (December 2018)	80% below 2005 levels by 2030 - Carbon-free electricity by 2050*

1 – Taking actions to support the objectives of the Paris climate agreement

2 – As of 2017, company had reduced CO₂ emissions rate to 108 lbs./MWh

3 – Applies to corporate emissions across all departments and operations

4 – Applies to electric energy portfolio for Montana

5 – Applies to carbon emissions from its natural gas customers' usage

* – Aspirational and/or dependent on advancements in technology and public policy

APPENDIX B



Edison Electric
INSTITUTE

Power by Association

August 26, 2019

Edward A. Boling
Associate Director for the National Environmental Policy Act
Council on Environmental Quality
730 Jackson Place, N.W.
Washington, D.C. 20503

Dear Associate Director Boling:

The Edison Electric Institute (EEI) appreciates the opportunity to provide the attached comments to the Council on Environmental Quality (CEQ) regarding the *Draft National Environmental Policy Act Guidance on Consideration of Greenhouse Gas Emissions* (Draft Guidance). 84 *Fed. Reg.* 30,097 (June 26, 2019). EEI is the association that represents all U.S. investor-owned electric companies. Our members provide electricity for about 220 million Americans, and operate in all 50 states and the District of Columbia. As a whole, the electric power industry supports more than seven million jobs in communities across the United States.

CEQ's Draft Guidance largely strikes an appropriate balance, encouraging agencies to use their expertise and experience to determine how and to what degree to analyze reasonably foreseeable potential direct and indirect effects of greenhouse gas (GHG) emissions; quantify such emissions when they are substantial; and provide well-reasoned explanations for their analyses and quantifications. CEQ should endeavor to finalize guidance that aids in the development of NEPA analyses that can survive judicial scrutiny, considering applicable judicial decisions, which will in turn allow projects requiring federal permits to move forward without undue delay.

The electric power industry is in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation is being driven by a wide range of factors, including: declining costs for natural gas and renewable energy resources, technological improvements, changing customer expectations, federal and state regulations and policies, and the increasing use of distributed energy resources. As a result, the mix of resources used to generate electricity in the United States has changed dramatically over the last decade and is increasingly clean.

These changes have had an impact on the sector's GHG emissions. As of the end of 2018, the electric power sector has reduced its carbon dioxide (CO₂) emissions by 27 percent compared with near peak levels in 2005—nearly the lowest level in 30 years. EEI's member companies have reduced their CO₂ emissions even more dramatically to approximately 37 percent below 2005 levels, based on an EEI analysis of industry emissions data. And these reductions will

continue: EEI's member companies are on a path to reduce CO₂ emissions nearly 50 percent by 2030 and potentially 80 percent by 2050 compared to 2005 levels while keeping electricity affordable and reliable.

The Draft Guidance also encourages agencies to use existing emissions estimates, including sector-wide estimates, to provide context for understanding the relative magnitude of a proposed action's GHG emissions. Such context is particularly important with respect to proposed actions that implicate electricity generation, transmission, and distribution, as the total emissions for the sector have decreased significant in recently years and will continue to decline. CEQ should ensure that any final guidance encourages agencies to consider the larger context of electricity sector GHG emissions where appropriate.

Questions may be directed to Eric Holdsworth (eholdsworth@eei.org, 202-508-5103) or Alex Bond (abond@eei.org, 202-508-5523).

Sincerely,

A handwritten signature in black ink, appearing to read "Eric Holdsworth", with a long horizontal line extending to the right.

Eric Holdsworth
Senior Director, Climate Programs

**COMMENTS OF THE EDISON ELECTRIC INSTITUTE
DRAFT NATIONAL ENVIRONMENTAL POLICY ACT GUIDANCE
ON CONSIDERATION OF GREENHOUSE GAS EMISSIONS**

DOCKET NO. CEQ-2019-0002

August 26, 2019

The Edison Electric Institute (EEI) appreciates this opportunity to submit comments on the *Draft National Environmental Policy Act Guidance on Consideration of Greenhouse Gas Emissions* issued recently by the Council on Environmental Quality (CEQ). 84 *Fed. Reg.* 30,097 (June 26, 2019)(Draft Guidance). The proposed guidance is intended to assist federal agencies in addressing greenhouse gas (GHG)¹ emissions in environmental assessments and impact statements conducted pursuant to the National Environmental Policy Act (NEPA). *See id.* Such guidance is critical to ensure that agencies can navigate the complexities and uncertainties related to assessing the impacts of direct and indirect GHG emissions for energy infrastructure projects in a manner that properly informs reasoned decision making, withstands judicial scrutiny, and respects the spirit of the statute. CEQ should consider the comments in response to this draft and provide final guidance expeditiously.

EEI is the association that represents all U.S. investor-owned electric companies. Our members provide electricity for about 220 million Americans and operate in all 50 states and the District of Columbia. As a whole, the electric power industry supports more than seven million jobs in communities across the United States. EEI member companies invest more than \$100 billion

¹ For purposes of this Draft Guidance, CEQ defines GHGs to include carbon dioxide (CO₂), methane, nitrous oxide, hydrofluorocarbons, perfluorocarbons, sulfur hexafluoride, and nitrogen trifluoride. *See* 84 *Fed. Reg.* at 30,097. These comments focus on NEPA assessments of GHG emissions from electric companies, the vast majority of which are CO₂ emissions.

dollars annually to make the energy grid smarter, cleaner, more dynamic, more flexible, and more secure in order to provide affordable and reliable electricity to customers.² Such projects frequently require federal permits and are therefore subject to NEPA reviews, which have increasingly become the focus of protracted litigation. Accordingly, EEI's member companies have a compelling interest in NEPA guidance that will assist agencies in addressing the complexities of assessing the direct and indirect impacts of GHG emissions associated with infrastructure projects while taking the requisite "hard look" at environmental consequences mandated by existing judicial precedent on the scope of NEPA reviews.

The Draft Guidance largely strikes an appropriate balance, encouraging agencies to use their expertise and experience to determine how and to what degree to analyze reasonably foreseeable potential direct and indirect effects of GHG emissions; quantify such emissions when they are substantial; and provide well-reasoned explanations for their analyses and quantifications. *See id.* at 30,098. CEQ should endeavor to finalize guidance that aids in the development of NEPA analyses that can survive judicial scrutiny, considering applicable judicial decisions, which will in turn allow projects requiring federal permits to move forward without undue delay.

The Draft Guidance encourages agencies to use existing emissions estimates, including sector-wide estimates, to provide context for understanding the relative magnitude of a proposed action's GHG emissions. *See id.* Such context is particularly important with respect to proposed actions that implicate electricity generation, transmission, and distribution, as the total emissions

² *See* EEI, Industry Data, Statistical Highlights: Capacity and Generation (2018), <http://www.eei.org/resourcesandmedia/industrydataanalysis/industrydata/Pages/default.aspx>.

for the sector have decreased significant in recently years and will continue to decline. CEQ should ensure that any final guidance encourages agencies to consider the larger context of electricity sector GHG emissions where appropriate.

I. The Draft Guidance Appropriately Balances The Requirements of NEPA With The Challenges of GHG Emissions Quantification and Clarifies That GHG Emissions Do Not Drive NEPA Analyses or Federal Decision-Making; CEQ Should Consider Providing Additional General Guidance.

This Draft Guidance marks CEQ's third effort to assist federal agencies as they engage in NEPA analyses that include consideration of GHG emissions. EEI provided comments in both 2010 and 2015 on the agency's previous proposals.³ In both instances, EEI encouraged CEQ to balance the information disclosure requirements of NEPA with the inherent difficulties of quantifying reasonably foreseeable direct and indirect GHG emissions related to a specific proposed action or project. Of particular concern to EEI members was the risk that agencies would engage in highly speculative emissions analyses, even when potential emissions might not be significant.

While these concerns remain relevant in the context of any GHG emissions analysis, they have been exacerbated by CEQ's previous draft guidance, which implied that increases in GHG emissions alone could drive not only the NEPA process, but also the federal decision-making process that NEPA serves. GHG emissions are clearly an important environmental issue, and considering such emissions is appropriate in the context of environmental reviews designed to ensure well-informed decision-making and comparison of alternatives. However, GHG emissions and potential climate impacts alone are not the only relevant or determinative factor in NEPA analyses. Other environmental, contextual, and statutory factors will be and must be

³ EEI's previous comments are attached here as Appendices A and B.

considered, too. The Draft Guidance strikes the right balance by encouraging agencies to assess and quantify GHG emissions when feasible and appropriate but also making clear that “NEPA analyses need not give greater consideration to potential effects from GHG emissions than to other potential effects on the human environment.” 84 *Fed. Reg.* at 30,098.

However, a goal of any final guidance from CEQ also should be to help agencies comply with applicable judicial decisions concerning the sufficiency of GHG emissions assessments undertaken to comply with NEPA. Legally insufficient NEPA analyses can lead to costly delays and investor uncertainty. Specifically, CEQ’s final guidance should help agencies navigate the various judicial decisions finding that failure to quantify GHG emissions from proposed actions and projects violates NEPA.⁴ It is critical to EEI’s members that any final guidance clarify that, while GHG emissions alone cannot and should not drive federal decision-making, agencies should take steps to quantify direct and indirect GHG emissions—and appropriately contextualize and caveat such quantifications—to satisfy NEPA’s information-forcing requirements.

Many legal decisions turn on the methods for and the extent to which agencies should consider the indirect and cumulative impacts associated with reasonably foreseeable direct or indirect

⁴ For example, several courts have found that agencies’ failure to quantify reasonably foreseeable GHG emissions render NEPA analysis insufficient. *See, e.g., Sierra Club v. FERC*, 867 F.3d 1357 (D.C. Cir. 2017); *WildEarth Guardians v. Zinke*, No. 1:16-cv-1724 (D.D. C. Mar. 19, 2019); *Indigenous Environmental Network v. U.S. Dept. of State*, No. 4:17-cv-00029 (D. Mont. Nov. 8, 2018).

GHG emissions.⁵ Legal decisions make clear that the quantification of such emissions and an assessment of their potential impacts is a very fact-specific endeavor, however, which turns not only on the proposed action, but also on the scope of the agency's authority and the information that it has available to it at the time of the analysis.⁶ Accordingly, if the goal is legally sufficient environmental analyses, it may be helpful for CEQ to provide agencies that may be required to engage in such analyses with general guidance on how to approach indirect and cumulative impacts for GHG emissions in the development of NEPA analyses. As discussed in more detail below, it is appropriate that CEQ encourage agencies to consider existing emissions estimates, including sector-wide estimates, when assessing indirect and cumulative GHG emissions impacts.⁷

The Draft Guidance appropriately recognizes that neither NEPA or CEQ's own implementing regulations require agencies to monetize costs and benefits of any proposed action or project when comparing alternatives, or to use the Social Cost of Carbon estimates. *See* 84 *Fed. Reg.* at 30,098-99. However, courts have required agencies to disclose the social costs of GHG emissions where they have also monetized economic benefits, such as royalty revenue and labor

⁵ NEPA requires that an agency consider the direct, indirect, and cumulative impacts of a its proposed actions and projects. *See* 40 C.F.R. § 1508.25(c).

⁶ *See, e.g., WildEarth Guardians*, n.4, *supra* (finding that the Bureau of Land Management (BLM) was required to consider in its NEPA analysis the reasonably foreseeable indirect emissions from oil and gas leasing but remanding to BLM to determine whether and how to attempt quantification of those emissions).

⁷ To this end, it also may be helpful for CEQ to provide examples of how Agencies could address whether emissions are significant for NEPA purposes.

income from fossil fuel leasing.⁸ Any final guidance should recognize these decisions and provide agencies with some rubric for identifying the threshold at which quantification or monetization of GHG emissions and impacts may be necessary for a full and balanced assessment of impacts under NEPA to help minimize the risk of future litigation on proposed infrastructure projects.

II. Electric Companies Continue To Lead the Clean Energy Transition And Reducing GHG Emissions; NEPA Guidance And Analyses Should Recognize These Reductions.

The electric power industry is in the middle of a profound, long-term transformation in how electricity is generated, transmitted, and used. This transformation is being driven by a wide range of factors, including: declining costs for natural gas and renewable energy resources, technological improvements, changing customer expectations, federal and state regulations and policies, and the increasing use of distributed energy resources. As a result, the mix of resources used to generate electricity in the United States has changed dramatically over the last decade and is increasingly clean. Natural gas surpassed coal as the main source of electricity generation in the United States for the third year in a row in 2018, with natural gas-based generation powering 34.9 percent of the country's electricity, compared to coal-based generation at 27.2 percent.⁹ Since 2014, more than half of the industry's investments in new electricity generation

⁸ See, e.g., *High Country Conservation Advocates v. USFS*, 52 F. Supp. 3d 1174 (D. Colo. 2014); *Montana Environmental Information Center v. Office of Surface Mining*, No. 9:15-cv-00106 (D. Mont. Nov. 3, 2017).

⁹ See Department of Energy, Energy Information Agency (EIA), *Electricity Explained: Electricity in the U.S.* (Apr. 2019), https://www.eia.gov/energyexplained/index.php?page=electricity_in_the_united_states. In April 2019, U.S. monthly electricity generation from renewable sources exceeded coal-fired generation for the first time based on data in EIA's *Electric Power Monthly*. Renewable sources provided 23 percent of total electricity generation to coal's 20 percent. According to EIA, this outcome reflects both seasonal factors as well as long-term increases in renewable generation and

have been in wind and solar generation resources,¹⁰ and more than one-third of American's electricity now comes from carbon-free resources, including nuclear power, hydropower, solar, and wind.¹¹ The trend of increasing renewable energy deployments will continue. EIA projects that the U.S. will add 72 gigawatts (GW) of new wind and solar capacity between 2018 and 2021 alone and that, long-term, demand for new electric generating capacity will be met by renewables and efficient natural gas as older coal-based and less-efficient natural gas-based generating units retire.¹²

These changes have had an impact on the sector's GHG emissions. As of the end of 2018, the electric power sector has reduced its CO₂ emissions by 27 percent compared with near peak levels in 2005—nearly the lowest level in 30 years. EEI's member companies have reduced their CO₂ emissions even more dramatically to approximately 37 percent below 2005 levels.¹³ And these reductions will continue: EEI's member companies are on a path to reduce CO₂ emissions

decreases in coal generation. EIA, *U.S. Electricity Generation from Renewables Surpassed Coal in April* (June 26, 2019), <https://www.eia.gov/todayinenergy/detail.php?id=39992>.

¹⁰ See EIA, *Nearly Half of Utility-Scale Capacity Installed in 2017 Came from Renewables* (Jan. 10, 2018), <https://www.eia.gov/todayinenergy/detail.php?id=34472>. See also EEI, *Industry Data, Statistical Highlights: Capacity and Generation* (2018), <http://www.eei.org/resourcesandmedia/industrydataanalysis/industrydata/Pages/default.aspx>.

¹¹ See EIA, *Electricity Explained: Electricity in the United States* (Apr. 2018), https://www.eia.gov/energyexplained/index.php?page=electricity_in_the_united_states.

¹² See EIA, *Annual Energy Outlook 2019: With Projections to 2050* (Jan. 24, 2019) at 94, 96, <https://www.eia.gov/outlooks/aeo/pdf/aeo2019.pdf>. While EIA notes in the *Annual Energy Outlook* (AEO) that the amount of renewable and natural gas-based generation deployed are dependent on the price of natural gas, this does not impact the expected closure of coal-based and other less efficient generation. See *id.* at 108.

¹³ Based on an EEI analysis of CO₂ emissions data from ABB Velocity Suite.

nearly 50 percent by 2030 and potentially 80 percent by 2050 compared to 2005 levels while keeping electricity affordable and reliable.¹⁴

Although not the subject of the Draft Guidance, other environmental benefits of this clean energy transition also include significant reductions in criteria pollutants, as well as acid gases and other hazardous air pollutants (HAPs), like mercury. As of 2018, emissions of sulfur dioxide (SO₂) and nitrogen oxides (NO_x) have been reduced by 92 and 84 percent, respectively, from 1990 levels.¹⁵ The industry has likewise reduced mercury emissions by approximately 86 percent from 2010 to 2017, and emissions of both acid gases and of total HAPs were reduced by 96 percent.¹⁶

The Draft Guidance encourages agencies to “reference local, regional, national, or sector-wide emissions estimates to provide context for understanding the relative magnitude of a proposed action’s GHG emissions.” 84 *Fed. Reg.* at 30,098. CEQ should include similar statements in the final guidance and encourage agencies to consider power sector reductions in both GHG and other emissions in NEPA assessments for projects or proposed actions that implicate electricity

¹⁴ Many EEI member companies have announced significant voluntary commitments to further reduce CO₂ emissions by 2030 and 2050, many of which aim to reduce emissions 80 percent below 2005 levels by 2050—these commitments are attached here as Appendix C. Further, EIA’s AEO 2019 base case projects that electric sector emissions will level off from 2020 to 2050, despite a projected 30 percent increase in generation over this period, as a result of decreases in the use of coal as a generating fuel. *See* EIA, n.10, *supra*, at 114. EIA’s base case only considers existing policies that are current laws and does not address the potential for future reductions as a result of new laws or industry commitments.

¹⁵ *See* U.S. Environmental Protection Agency (EPA), Power Plant Emission Trends, <https://www.epa.gov/airmarkets/power-plant-emission-trends>.

¹⁶ *See* EPA, *National Emission Standards for Hazardous Air Pollutants: Coal- and Oil-Fired Electric Utility Steam Generating Units—Reconsideration of Supplemental Finding and Residual Risk and Technology Review*, 84 *Fed. Reg.* 2,670, 2,689 (Feb. 7, 2019).

generation, transmission, or distribution. While there may be concerns about project- or action-specific efforts to quantify GHG emissions, EIA's careful and well-substantiated analyses unquestionably should satisfy NEPA's information-forcing objectives.¹⁷

This kind of sector-wide contextualization is particularly important when agencies must undertake NEPA GHG analyses for specific energy projects. GHGs, unlike other pollutants, are a well-mixed pollutant, meaning that the amount measured in the atmosphere is roughly the same all over the world, regardless of the location of any particular source of emissions.¹⁸ Moreover, it can be hard to accurately estimate project-specific indirect and cumulative GHG emissions impacts. Accordingly, a sector-wide assessment of total GHG emissions and emissions trends can help federal agencies and interested stakeholders better understand the potential direct, indirect, and cumulative emissions impacts of a proposed action. For example, sector-wide assessments can help demonstrate that a particular project—like a new electric transmissions line or natural gas pipeline—actually serves to reduce indirect emissions by allowing electric companies to integrate more renewable resources or shift from higher-emitting coal to lower-emitting natural gas as a fuel.¹⁹

¹⁷ EIA is but one example of rigorous sector-wide emissions analysis. Agencies should be encouraged to consider all such sources as they consider GHG emissions in the NEPA context, consistent with NEPA's information-forcing objectives.

¹⁸ See EPA, *Greenhouse Gas Emissions, Overview of Greenhouse Gas Emissions*, <https://www.epa.gov/ghgemissions/overview-greenhouse-gases>.

¹⁹ Similarly, the appropriate consideration of mitigation measures can help contextualize emissions. Agencies should be encouraged to consider such measures in NEPA analyses if relevant.

Importantly, where direct and indirect GHG emissions are difficult to quantify, sector-wide emissions assessments can provide helpful context that allows agencies to provide a qualitative assessment of the potential impacts of a proposed permit or action. The effects of GHG emissions from any particular project on future warming trends cannot be predicted precisely—especially at the local level—with current scientific tools.²⁰ However, reliable sector-wide assessments can help inform both agency decision-making and interested stakeholders consistent with the requirements and objectives of NEPA. Any final guidance should therefore emphasize that agencies could consider sector-wide emissions estimates when assessing the potential direct and indirect emissions from proposed actions and projects.

²⁰ IPCC, 2014: *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* [Core Writing Team, R.K. Pachauri and L.A. Meyer (eds.)]. IPCC, Geneva, Switzerland, 151 pp.

APPENDIX C



**Edison Electric
Institute**

Quinlan J. Shea, III
Vice President, Environment

March 25, 2015

Council on Environmental Quality
Attn: Horst Greczmiel
722 Jackson Place, NW
Washington, DC 20503

Submitted by e-mail to: GCC.guidance@ceq.eop.gov

Re: Revised draft CEQ NEPA Guidance on Consideration of Greenhouse Gas
Emissions and Effects of Climate Change, 79 Fed. Reg. 77802 (Dec. 24, 2015)

Dear Mr. Greczmiel:

The Edison Electric Institute (EEI) respectfully submits these comments in response to the above-referenced revised draft guidance, which is intended to provide federal agencies assistance and guidance in examining the potential effects of proposed actions and alternatives on greenhouse gas (GHG) emissions and the potential effects of climate change on the environmental consequences of proposed actions and alternatives when conducting environmental reviews under the National Environmental Policy Act (NEPA).

EEI is the association of U.S. investor-owned electric companies, with international affiliates and industry associates worldwide. Our members provide electricity for 220 million Americans, operate in all 50 states and the District of Columbia, and directly employ more than 500,000 workers. With more than \$90 billion in annual capital expenditures, the electric power industry also is responsible for millions of additional jobs outside of our direct operations. Reliable, affordable, and sustainable electricity powers the economy and enhances the lives of all Americans.

EEI's members rely on electricity generating, transmission, distribution and related facilities operating together in a complex, highly interrelated grid to provide the electricity on which agencies, businesses, cities and residents rely nationwide. Some of these facilities produce GHG emissions, and the environmental effects of some may be affected by changes in climate. Furthermore, some of the facilities require federal permits or other approvals, such as federal land use authorizations, that can trigger NEPA reviews. Because EEI members will be affected directly by any GHG-climate NEPA guidance, we have a direct interest in the draft guidance CEQ has proposed for public comment.

EEI submitted previously the enclosed extensive comments on CEQ's 2010 draft NEPA GHG-climate guidance. We raised a number of substantive concerns, especially regarding the need to avoid speculative NEPA analyses given the difficulty of predicting effects from emissions from any specific facility, as well as the limited contributions that individual facilities and actions generally make to global GHG emissions and concentrations. While CEQ has recognized and addressed some of EEI's earlier concerns in developing the current revised draft guidance, many remain applicable; hence, we encourage CEQ to reconsider EEI's earlier comments.

An important difference between the revised draft guidance and the 2010 draft guidance is the context in which they were proposed. The revised draft guidance follows the issuance of the proposed Clean Power Plan (CPP), the Environmental Protection Agency's (EPA's) guidelines for reducing emissions from existing electric generating units. Compliance with these guidelines will require dramatic changes to the way that electricity is generated, transmitted and consumed. To facilitate these changes, new electric generating units (EGUs) and new and expanded electric transmission lines and natural gas pipelines will be required. Many of these may be subject to NEPA reviews to obtain siting and other permits. As noted in the revised draft guidance, NEPA requires agencies to consider the extent to which proposed actions and their alternatives contribute GHGs. The final guidance should make clear that agencies also should consider the extent to which new EGUs, transmission lines and natural gas pipelines proposed to help states comply with their CPP goals contribute toward the reduction of overall emissions from the power sector. Important elements of state CPP plans should not be delayed or rejected merely because they may lead to *de minimis* increases in near-term GHG emissions if they are part of larger efforts to move to cleaner sources of electricity.

EEI appreciates several elements of CEQ's current draft guidance, including:

- Not proposing to change the basic framework for NEPA analysis set out in CEQ's current regulations, but instead attempting to provide clarity in how GHGs and climate change can be considered within that framework.
- Proposing to continue giving federal agencies significant discretion in determining when, whether and how to incorporate GHGs and climate change effects into their NEPA reviews of proposed actions.
- Referring several times to the "rule of reason" and the "concept of proportionality" that agencies should apply in undertaking NEPA reviews, including in determining the scope of reviews and effects, alternatives and potential mitigation measures.
- Instructing agencies to explore only reasonably foreseeable effects.
- Clarifying that cumulative GHG emissions—any individual proposed action's GHG emissions together with all others worldwide—alone do not require an environmental impact statement (EIS).

These limitations and directions should help reduce the risk of federal agencies conducting NEPA analysis of GHGs and climate effects out of proportion to their importance in the context of a given proposed action.

EEI nonetheless remains concerned that CEQ continues to call on federal agencies to engage in highly speculative analyses that will create substantial additional burden, cost, delay and uncertainty—both for projects that have a *de minimis* impact on total global GHG concentrations and for agencies and the regulated community, including federal permittees—without producing clear environmental benefits. We also remain concerned that, if adopted as final, the guidance would invite litigation over whether agencies have properly implemented it. Ironically, this would serve merely to exacerbate the negative effects of the draft guidance and to unnecessarily and inappropriately delay important projects, including those that reduce total emissions because they replace higher emitting activities.

At a minimum, to address these and other concerns raised in our 2010 comments, EEI encourages CEQ to make further changes to the draft guidance before issuing it as final, including:

- Specifying early in the guidance that GHG emissions and any corresponding climate change effects are but one piece of information that agencies will want to consider in deciding whether to proceed with a proposed action or alternative; and, if so, what reasonable mitigation measures should be included. For example, agencies typically make decisions on land and resource use after taking into account a large array of factors, including the statutory responsibility to encourage multiple uses of federal lands, the economic and energy benefits of a proposed action, and the overall environmental benefits and effects of the proposed action. Thus, GHG emissions and climate effects should not drive the NEPA analysis, but should be considered as one component. As EEI noted in its 2010 comments, individual proposed actions alone are not a proximate cause of climate change, and local climate change is not understood well enough to predict its effects on the environmental impacts of a project. CEQ should recognize this explicitly and avoid overstating the role of project-specific GHGs and climate change effects in the NEPA analyses. Again, in this regard, EEI strongly supports the statement in the draft guidance that cumulative GHG emissions alone do not require an EIS. Otherwise, every proposed action would be driven inappropriately to an EIS by GHG emissions that are a miniscule contribution to the worldwide total and do not proximately cause any climate change effects.
- Reminding agencies they only need to consider alternatives to a proposed action that are consistent with the purpose of and need for the proposed action. While the current draft guidance mentions this point, it gets lost in the overall discussion of ensuring that GHG emissions and climate effects on consequences of a proposed action are considered, including through exploration of alternatives that would reduce those emissions or effects.

- Recognizing that some proposed actions actually will reduce GHG emissions directly or indirectly, and so should receive positive credit in the NEPA analysis for those beneficial effects. Thus, CEQ's final guidance should expressly direct agencies to consider the net environmental effects of a proposed project, including a project's beneficial effects on GHG emissions. Such recognition is particularly important in the context of EPA's proposed CPP for the many projects that the electric and natural gas industry will undertake to implement this plan. For example, a base load natural gas-based power unit that replaces a coal-based unit and thereby reduces net emissions should be evaluated as creating a net reduction in GHG emissions and should not require further analysis of GHG impacts in the EIS.
- Raising the 25,000 metric ton per year benchmark (below which an agency "generally" need not discuss GHGs and climate effects "unless" data are available) to at least 100,000 metric tons per year, and removing the "generally" qualifier so GHGs and climate effects do not need to be discussed below the threshold. The higher number would conform to EPA's threshold for permitting GHG emissions from new facilities, thereby putting NEPA review of GHG and climate aspects of proposed new actions on par with the facilities covered by EPA permitting requirements, and removing "generally" would avoid confusion and review of truly *de minimis* emissions.
- Clarifying that, even above this threshold, consideration of GHG emissions and climate effects on a proposed action's effects may not be warranted, and stating explicitly in the guidance (as CEQ does in the preamble at 79 *Fed. Reg.* 77807) that the threshold does not in any way indicate "significance" for purposes of deciding whether to prepare an EIS. NEPA requires federal agencies to consider only the environmental impacts *proximately caused* by their actions. *Metropolitan Edison Co. v. People Against Nuclear Energy* ("PANE"), 460 U.S. 766, 774 (1983); *see also Dep't of Trans. V. Public Citizen*, 541 U.S. 752 (2004) (citing PANE). In particular, GHG emissions cannot be linked as a proximate cause of specific climate change effects, and agencies should be instructed not to do so in their NEPA analyses. Agencies should continue to decide what environmental effects to consider and whether to prepare an EIS using the existing factors set out in CEQ's NEPA regulations. This must be made very clear in the final guidance to avoid causing confusion for agencies, the regulated community and the courts.
- Instructing agencies not to discuss the links between GHG emissions of a proposed action and climate change effects in the vicinity of the proposed action or worldwide, because any such discussion would be inherently speculative. As CEQ has recognized, climate change is a global phenomenon, where worldwide natural and human-caused GHG emissions together may produce certain predicted climate effects. GHG emissions from particular actions are tiny in comparison to worldwide emissions. The effects of worldwide emissions on climate change cannot currently be predicted with necessary accuracy, and the effects of individual

emissions on local and regional climate are even more difficult, if not impossible, to estimate meaningfully with current scientific tools. At most, CEQ should direct agencies to disclose GHG emissions from both proposed actions and alternatives, but without engaging in speculation as to the specific ensuing effects of those emissions.

- Instructing agencies to focus on foreseeable climate change effects directly resulting from the proposed action, not “life-cycle” or “downstream” emissions, to avoid speculation as to effects that are remote in time and geographically from the proposed action, and that are likely to occur with or without the proposed action (e.g., increases in natural gas production likely to occur with or without natural gas-based electricity generation). We do not read the current draft guidance as requiring life-cycle analysis, and we note that the preamble to the draft guidance suggests that this is not CEQ’s intent. *See 79 Fed. Reg.* at 77805. Yet, the draft guidance’s discussion of “upstream” and “downstream” clouds the issue.
- Clarifying that NEPA focuses on the environmental effects of a proposed action, but then as a matter of sound decision-making agencies can take into account any benefits provided by a proposed action that increase the resiliency of critical assets to the effects of climate change.
- Clarifying that the environmental baseline for NEPA reviews remains the state of the environment as it currently exists, not a moving target that may change in the future with climate change, to avoid creating additional uncertainty in conducting analyses. The current draft guidance suggests that agencies should consider anticipated changes in climate as part of their NEPA analysis, but that is not possible with a reasonable degree of certainty, and especially so at a local level for a particular activity.
- Clarifying that agencies need not examine indirect effects of a proposed action if those effects are speculative or remote, as most indirect effects would be in the GHG and climate change context.
- Deleting the draft guideline’s directive at section B, *79 Fed. Reg.* 77827, to use the Federal social cost of carbon metric to evaluate and monetize costs and benefits in NEPA reviews. As the draft guidance recognizes, that metric was developed for use in the regulatory analysis context, not to quantify actual costs or benefits of GHG emissions in the context of a particular agency action or NEPA review of such action. Furthermore, the metric has raised a number of concerns, including that it was developed through a closed rather than transparent process, its assumptions are unclear, and it uses arbitrary time frames, discount rates and input data. Moreover, use of the metric in the context of individual agency actions suggests that the actions have proximate cause on climate change and effects that warrant quantification, when as discussed above, that is not true.

- Eliminating the reference to “constructing transmission towers” from among the list of examples of projects that could benefit from programmatic or broad-based NEPA reviews in the GHG and climate change context. *See 79 Fed. Reg. 77830.*
Transmission lines are non-emitting facilities and do not proximately cause any changes in emissions from generating facilities. As explained in more detail in EEI’s enclosed 2010 comments (pp. 14-16), transmission lines are single elements in a highly integrated grid that support the delivery of electricity from generation sources to customers. Unless a new line was dedicated to serve particular new generation capacity—a relatively rare occurrence given the interconnected nature of the grid—construction of transmission lines do not cause any changes in generation. The extent to which any given generation facility is used to generate electricity depends on fluctuations in demand driven by factors such as the time of day, weather and fuel prices, and not on the presence or absence of any particular transmission line.
- Not requiring agencies to engage in climate change modeling or data gathering, but at most to rely on existing information to avoid inappropriately increasing the burden of conducting a NEPA analysis far out of proportion to the individual proposed action. The current draft guidance appears consistent with this point, yet it calls on agencies to gather additional data if the cost of doing so is reasonable.
- Clarifying that agencies are not required to adopt mitigation measures as part of NEPA analyses. NEPA is a procedural statute, not one that imposes particular substantive requirements. While agencies should consider mitigation requirements as appropriate, the guidance should not suggest that their NEPA analyses must specify such requirements, which remain within the discretion of the action agency.
- Specifying that the guidance, once final, will apply prospectively and should not be applied to ongoing projects. Because a NEPA analysis can be a long and sometimes costly process that expends considerable resources—both for agencies conducting reviews and project owners with permits subject to these reviews—a requirement to apply new guidance to projects already underway would be more costly and infuse more uncertainty and potential delay into the NEPA permitting process. CEQ should clarify that final guidance will only apply to new NEPA reviews that have not undergone the scoping process.

EEI appreciates the opportunity to submit these comments. If you have any questions or need additional information, please do not hesitate to contact me, Henri Bartholomot (hbartholomot@eei.org or 202-508-5622), Emily Fisher (efisher@eei.org or 202-508-5616), Eric Holdsworth (eholdsworth@eei.org or 202-508-5103) or Rick Loughery (rloughery@eei.org or 202-508-5647).

Sincerely,



Quinlan J. Shea, III

Enclosure

APPENDIX D



**EDISON ELECTRIC
INSTITUTE**

QUINLAN J. SHEA, III
Executive Director, Environment

May 24, 2010

Ted Boling, Esq.
Senior Counsel
Council on Environmental Quality
722 Jackson Place, NW
Washington, DC 20503

Submitted by e-mail to: CE.guidance@ceq.eop.gov

Re: Draft CEQ NEPA Guidance on Consideration of Climate Change
and Greenhouse Gases, 75 Fed. Reg. 8046 (February 23, 2010)

Dear Mr. Boling:

The Edison Electric Institute is pleased to submit the attached comments to the Council on Environmental Quality on the above-referenced "Draft NEPA Guidance on Consideration of the Effects of Climate Change and Greenhouse Gas Emissions," dated February 18, 2010.

EEI appreciates the opportunity to submit these comments. If you have any questions about them, or need additional information, please contact me, Edward Comer at (ecomer@eei.org; 202-508-5615), William Fang at (bfang@eei.org; 202-508-5617) or Henri Bartholomot at (hbartholomot@eei.org; 202- 508-5622).

Respectfully submitted,

A handwritten signature in black ink, appearing to read "Quinlan J. Shea", with a stylized flourish at the end.

Quinlan Shea
Executive Director, Environment

QS:cfb
Attachment

**COMMENTS OF THE EDISON ELECTRIC INSTITUTE
ON THE COUNCIL ON ENVIRONMENTAL QUALITY'S
"DRAFT NEPA GUIDANCE ON CONSIDERATION OF THE EFFECTS OF
CLIMATE CHANGE AND GREENHOUSE GAS EMISSIONS"**

May 24, 2010

The Edison Electric Institute (EEI) is pleased to submit these comments to the Council on Environmental Quality (CEQ or Council) on its "Draft NEPA Guidance on Consideration of the Effects of Climate Change and Greenhouse Gas Emissions," dated February 18, 2010. CEQ announced that the guidance was available, and invited comments by May 24, 2010, in a notice published at 75 *Fed. Reg.* 8046 (Feb. 23, 2010).

In that notice, CEQ specifically invited comments on "when and how Federal agencies must consider the impacts of proposed Federal actions on global climate change, as well as the expected environmental effects from climate change that may be relevant to the design of the proposed Federal action." EEI addresses both of these issues in our comments.

EEI is the association of shareholder-owned electric utilities in the U.S. Our members serve 95 percent of the ultimate customers in the shareholder-owned segment of the industry, and they represent approximately 70 percent of the nation's electric power industry. Much of the electricity required for homes and businesses is generated through means that produce carbon dioxide (CO₂), and utility operations may result in emissions of other greenhouse gases (GHGs). In addition, EEI members are affected directly as permittees, and indirectly through their reliance on federal agency resource decisions, by federal agency implementation of the National

Environmental Policy Act of 1969, 42 U.S.C. § 4321 *et seq.* (NEPA). Therefore, EEI has a direct interest in the draft guidance.

I. Overview

EEI is submitting these comments to CEQ because the power sector has a major interest in ensuring that environmental reviews of electric generation, transmission, and distribution projects and other such critical infrastructure – including fuel resources – are appropriate in nature, scope and content. We also seek to ensure that the environmental review process for such facilities is as streamlined and efficient as possible in order to avoid unnecessary delays and increased costs to power sector customers in obtaining necessary agency decisions as to the facilities.

EEI supports comprehensive, economy-wide climate change legislation. Comprehensive legislation is simply a better fit for addressing climate change than traditional environmental regulation under statutes such as the Clean Air Act (CAA), Clean Water Act, Endangered Species Act (ESA) and NEPA. Well-designed legislation can achieve significant GHG emissions reductions – for example, by promoting enhanced energy efficiency, lower GHG-emitting technology, renewable energy sources, advanced coal technologies coupled with carbon capture and storage (CCS), and other critical technologies and measures – while helping to maintain both the reliability and affordability of the nation’s electric energy supply.

We fully support the intent of NEPA to ensure that federal agencies consider significant environmental effects of their decisions on the human environment. At the same time, the NEPA

review process and decisions based on it need to be reasonable and timely in order to ensure the best use of limited federal agency, regulated community and public resources.

Agency reviews under NEPA already provide a comprehensive and sound review of proposed federal actions and their environmental impacts. Therefore, in order to avoid confusion, duplication, litigation and delay, CEQ guidance should be properly tailored to help to ensure consistency across federal agencies and reasonable, streamlined implementation of NEPA. To achieve these goals, guidance for considering climate change and GHGs in the NEPA context must be crafted with great care. GHG emissions are unlike traditional pollutants regulated under various environmental statutes because there is little (if any) relationship between GHG emissions from a project in a particular location and the possible environmental effects of climate change in that location. Instead, it is the total global accumulation of GHG emissions over a long period of time that matters. In addition, future climate conditions cannot be predicted or quantified with certainty at a global level, much less at a national, regional or local level. Furthermore, given the current state of general circulation models (GCMs), the science cannot definitively predict climate change effects resulting from GHG emissions at individual facilities at any level – national, regional or local. This makes the process of considering the effects of GHGs and climate change from individual project proposals inherently speculative. While CEQ has recognized this in part in the draft guidance, the final guidance should more fully reflect this basic reality in order to avoid encouraging federal agencies to spend time and funds on GHG and climate analyses that, in the end, would not meaningfully inform the agency decisions or even worse, could provide confusing or misleading information.

In addition, CEQ's guidance should reflect more clearly that CEQ's proposed reference level of 25,000 metric tons per year of CO₂ equivalent (CO₂e) does not constitute a basis for requiring more detailed environmental analysis of a particular activity. As we explain below, the 25,000-ton indicator was developed to collect emissions data, not as a regulatory benchmark. In fact, 25,000 tons represents only 5/100,000th of 1 percent (0.0000005) of the 49 billion tons of global GHG emissions in 2004. EPA's so-called "tailoring" rule, issued on May 13, 2010, will apply 75,000 and 100,000 tons per year as the minimum levels for triggering Prevention of Significant Deterioration (PSD) permits under the CAA. While it may be appropriate to report projected GHG emissions at such levels in an environmental impact statement (EIS) in order to inform the decision-maker and the public, even a 100,000-tons-per-year project will have an imperceptible impact upon climate, and therefore agencies should not be tasked with engaging in any further analysis that would be largely speculation.

While EEI appreciates CEQ's desire to provide guidance on how to consider climate change and GHGs in NEPA reviews, we encourage the Council to refine the proposed guidance to provide greater clarity and more reasonable recommendations on how to accomplish this. At most, the guidance should require agencies to include information about anticipated GHG emissions in environmental analyses of major projects. The guidance should not require agencies to engage in qualitative discussion of the impacts of such emissions upon climate because any such changes are far too speculative to predict. Nor should agencies be required to conduct detailed environmental analyses, including EISs, simply because of an activity's GHG emissions. Instead, publication of projected annual emissions levels should be sufficient under NEPA analysis.

II. Because There Is No Substantive Basis To Draw A Link Between A Particular Project's GHG Emissions And Climate Change, Discussion Of The Effects Of Such Emissions On Climate Change And Alternatives For Reducing Those Emissions Would Be Speculative And Unproductive.

In the draft guidance, CEQ correctly concedes that “it is not currently useful” to attempt to make a causal link between climate change, or the environmental impacts of such change, and a particular project or its emissions because “such direct linkage is difficult to isolate and to understand” (p. 3). CEQ also acknowledges the view of the Environmental Protection Agency (EPA) that it is not currently “possible to quantify with great specificity (i.e. geographic) the various health effects from climate change.”¹ *Id.* at 11. In addition, CEQ recognizes that the Supreme Court has stated that the obligation of an agency to discuss particular effects turns on “a reasonably close causal relationship between the environmental effect and the alleged cause.” *Id.* at 7, quoting *Dep’t of Transp. v. Public Citizen*, 541 U.S. 752, 767 (2004)).

Because there is no such demonstrable “causal relationship” between an individual activity’s GHG emissions and specific climate change impacts, there is no meaningful basis for requiring agencies to engage in a speculative discussion of such a link in the NEPA context. Yet the draft

¹ In the proceeding where EPA made endangerment findings under the CAA, the EPA Administrator said:

The Administrator recognizes that human-induced climate change has the potential to be far-reaching and multi-dimensional, and in light of existing knowledge, that not all risks and potential impacts can be quantified or characterized with uniform metrics. There is variety not only in the nature and potential magnitude of risks and impacts, but also in our ability to characterize, quantify and project such impacts into the future.

74 *Fed. Reg.* 66,496, 66,497 (2009) (emphasis added).

guidance says that it would be appropriate, in a federal agency's analysis of an activity's direct effects under NEPA, for the agency to quantify the activity's GHG emissions, to discuss "qualitatively" the link between those emissions and climate change, and to consider measures to reduce the emissions, including consideration of reasonable alternatives. *Id.* at 3.

While EEI agrees that agencies can provide some reasonable estimates of GHG emissions anticipated from individual activities, we strongly disagree that agencies can or should be expected to discuss in their NEPA analyses the impact of an activity's GHG emissions on climate change, or the impacts of climate change on the activity, because such discussions would be speculative and thus unproductive and unhelpful to agency decision-making.

A. Courts Agree that under NEPA, Agencies Need Not Analyze Effects Absent a Causal Link and Need Not Engage in Speculative Alternatives Analysis.

The courts have long applied a "rule of reason" to NEPA analysis, which holds that agencies are not required to engage in speculation or "crystal ball inquiry." In *National Resources Defense Council, Inc. v. Morton*, 458 F.2d 827 (D.C. Cir. 1972), the court examined the need to consider reasonable alternatives in an EIS in the context of an oil and gas lease sale under the Outer Continental Shelf Lands Act and concluded:

[T]he requirement in NEPA of discussion as to reasonable alternatives does not require "crystal ball" inquiry. ... The statute must be construed in the light of reason if it is not to demand what is, fairly speaking, not meaningfully possible, given the obvious, that the resources of energy and research — and time — available to meet the Nation's needs are not infinite. ... In the last analysis, the requirement as to alternatives is subject to a construction of reasonableness, and we say this with full awareness that this approach necessarily has both strengths and weaknesses.

458 F.2d at 837-38 (emphasis added). CEQ's current regulations adopt these same principles.

40 C.F.R. §§ 1500.4(f)-(g), 1501.7 and 1508.25. The draft guidance states that "[a]gencies apply

the rule of reason” on the usefulness of information regarding the “proposal, its alternatives, and mitigations options” (p. 5). It also appropriately refers to *Public Citizen*, supra, where the Supreme Court held:

[I]nherent in NEPA and its implementing regulations is a “rule of reason,” which ensures that agencies determine whether and to what extent to prepare an EIS based on the usefulness of any new potential information to the decisionmaking process. See *Marsh*, 490 U.S., at 373-374, 109 S.Ct. 1851. Where the preparation of an EIS would serve “no purpose” in light of NEPA’s regulatory scheme as a whole, no rule of reason worthy of that title would require an agency to prepare an EIS. See *Aberdeen & Rockfish R. Co. v. Students Challenging Regulatory Agency Procedures (SCRAP)*, 422 U.S. 289, 325, 95 S.Ct. 2336, 45 L.Ed.2d 191 (1975); see also 40 CFR §§ 1500.1(b)-(c) (2003).

541 U.S. at 767-68.

Federal agencies should not be encouraged to engage in “qualitative” analysis of the effects of GHG emissions or alternatives for reducing GHGs if that would require a “crystal ball inquiry.” The “rule of reason” must apply when effects from an individual project proposal cannot be readily ascertained and the value (if any) of the alternatives is remote and speculative.

B. There Is No Demonstrable Causal Link between an Activity’s GHG Emissions and Climate Change, and Alternatives Analysis in this Context is Inherently Speculative.

The draft guidance accurately states that “it is not currently useful for the NEPA analysis to attempt to link specific climatological changes, or the environmental impacts thereof, to the particular project or emissions, as such direct linkage is difficult to isolate and understand” (p. 3). This position is consistent with the positions taken by other federal agencies.

In a May 14, 2008, memorandum, the Department of the Interior (DOI) stated, “It is currently beyond the scope of existing science to identify a specific source of CO₂ emissions and designate

it as the cause of specific climate impacts at an exact location.” Mark D. Myers, Director, U.S.

Geological Survey re: “The Challenges of Linking Carbon Emissions, Atmospheric Greenhouse Gas Concentrations, Global Warming, and Consequential Impacts” (emphasis added).

Subsequently, in an October 3, 2008, letter to the Director of the U.S. Fish and Wildlife Service and the Director of the Office of Protected Resources, National Marine Fisheries Service regarding the “Endangered Species Act and GHG Emitting Activities,” EPA cited the DOI memorandum and agreed that current scientific analysis lacked the “tools” to make the “linkage” referred to by CEQ. EPA’s Principal Deputy Assistant Administrator, Office of Air and Radiation stated:

To date, research on how emissions of CO₂ and other GHGs influence global climate change and associated effects has focused on the overall impact of emissions from aggregate regional or global sources. This is primarily because GHG emissions from single sources are small relative to aggregate emissions, and GHGs, once emitted from a given source, become well mixed in the global atmosphere and have a long atmospheric lifetime. The climate change research community has not yet developed tools specifically intended for evaluating or quantifying end-point impacts attributable to the emissions of GHGs from a single source, and we are not aware of any scientific literature to draw from regarding the climate effects of individual, facility-level GHG emissions.

p. 4 (emphasis added).

In short, until the science of GCMs improves, no one will be able to model with any certainty or reliably predict the effects of a particular project’s GHG emissions on climate change. In recognition of the fact that such modeling would be an exercise in sheer speculation, the Obama Administration is initiating an almost \$50-million, five-year research effort by the National Science Foundation and the Agriculture and Energy Departments “aimed at producing new, high-resolution computer models” with the goals of producing “reliable predictions of climate

change at regional decadal scales, the resulting impacts, and the potential adaptation of living systems to these impacts.”²

Climate change is a global phenomenon caused by cumulative global GHG emissions over long periods of time, and is not a national, regional or local phenomenon linked to individual project emissions.³ Thus, the draft guidance calls climate change a “global problem that results from global GHG emissions,” while appropriately noting that “there are more sources and actions emitting GHGs (in terms of both absolute numbers and types) than are typically encountered when evaluating the emissions of other pollutants” (p. 2). Indeed, the Economic Report of the President to Congress (February 2010) observes, “Developing countries such as China and India are responsible for a growing proportion of emissions because of their heavy reliance on carbon-intensive fuels, such as coal . . .” (p. 255).

Both the Administration and Congress have recognized that addressing climate change will require concerted action globally, through international treaties and national actions, including enactment of economy-wide congressional legislation in the U.S., that together address the matter comprehensively and effectively. Furthermore, there are no GHG “hot spots,” such as occurs with some traditional air pollutants. GHG emissions do not have localized effects. For

² Energy Daily (March 25, 2010).

³ The Energy Information Administration’s (EIA) International Energy Outlook 2009, Chapter 8, titled “Energy-Related Carbon Dioxide Emissions,” states, “Because anthropogenic emissions of carbon dioxide result primarily from the combustion of fossil fuels, world energy use continues to be at the center of the climate debate” (p. 109).

these reasons, EPA has not targeted GHG emissions for regulation under the CAA in any regulation aimed at improving air quality in specific localities, states or regions of the U.S.

Quantitatively, the draft guidance states that this “problem is much more the result of numerous and varied sources, each of which might seem to make a relatively small addition to global atmospheric GHG concentrations” (p. 2). While the draft guidance briefly recognizes the global context of GHG emissions and climate change, CEQ appears to be attempting to mitigate the impacts of climate change at a local project level under a statute that is ill-suited to address the global nature of GHG emissions. This is precisely why Congress is attempting to legislate an appropriate response to the issue rather than leaving it to EPA under the CAA to regulate GHG emissions in inappropriate, costly and awkward ways.

Furthermore, the nature of climate change raises the problem of “leakage.” If NEPA analysis of GHG emissions and climate change should stall or stop any specific project, this could in many cases simply shift the emissions overseas, perhaps even increasing net global emissions.⁴ The energy-related CO₂ emissions of non-Organization of Economic Cooperation and Development (OECD) nations surpassed the CO₂ emissions of OECD countries in 2006, and by 2030 non-

⁴ In its April 28, 2010, report to Congress titled EPA Regulation of Greenhouse Gases: Congressional Responses and Options, the Congressional Research Service concluded:

The differences are of scale and of degree. Greenhouse gases are global pollutants to a greater extent than most of the pollutants previously regulated under the [Clean Air] Act; reductions in U.S. emissions without simultaneous reductions by other countries may somewhat diminish but will not solve the problems the emissions cause.

pp. 12-13 (footnote omitted and emphasis added).

OECD emissions are projected to be 73 percent greater than those of OECD nations.⁵ Since the already minuscule contribution of any specific federal project's GHG emissions to cumulative global emissions will only become smaller and smaller in the future, it would counterproductive to shift those emissions overseas.

C. 25,000 Metric Tons Per Year Should Not Be a Trigger for Quantitative Analysis.

The draft guidance chooses as a trigger 25,000 metric tons or more of CO₂e GHG emissions and indicates that “if a proposed action would be reasonably anticipated to cause direct emissions” in that quantity annually, “agencies should consider this as an indicator that a quantitative and qualitative assessment may be meaningful to decision makers and the public” (p. 1) (emphasis added). The draft guidance adds that in the case of “long-term actions that have annual direct emissions of less than” such quantity, CEQ “encourages” the agencies “to consider whether the action’s long-term emissions should receive similar analysis.” *Id.* at 1-2.

The draft guidance then states that CEQ is “not” proposing this metric “as an indicator of a threshold of significant effects, but rather as an indicator of a minimum level” of emissions “that may warrant some description in the appropriate NEPA analysis” of agency actions “involving direct” GHG emissions. *Id.* at 2. The draft guidance also notes that this number now “serves as a minimum standard for reporting emissions” that has been recently promulgated by EPA.⁶ *Id.* at 3. The draft guidance contends that the 25,000-metric-tons figure “may provide a useful, presumptive, threshold for discussion and disclosure of GHG emissions because it has been used

⁵ See EIA, *supra* n. 3, at 6 (Fig. 9), 109 (Fig. 80).

⁶ We disagree with CEQ’s characterization of the EPA reporting rules as being issued under the CAA. In fact the initial reporting rule was issued under the Omnibus Appropriations Act of FY08.

and proposed” by EPA and “because it provides comprehensive coverage of emissions . . . , thereby creating an important data set useful in quantitative analyses of GHG policies, programs and regulations” and, as such, this “rationale is pertinent to the presentation of NEPA analysis as well.” *Id.* at 3 n. 2.

While it is reasonable for CEQ to attempt to provide a benchmark level above which agencies should examine GHG emissions from a proposed activity in undertaking NEPA analyses, we are concerned that – notwithstanding CEQ comments in the draft guidance to the contrary – agencies would improperly treat this low level of emissions as a “significant effect” threshold for NEPA purposes.

First, CEQ has failed to provide any scientific justification for the 25,000-ton indicator. If science is truly to be the foundation for the Administration’s environmental policy, there should be a principled rationale for such a number.

Second, an individual project’s GHG emissions – even for projects with relatively large emissions – will be infinitesimal compared to worldwide GHG emissions, and more importantly the individual project’s emissions will not have a demonstrable effect on climate change. This is clearly true for projects emitting 25,000 tons of GHG emissions annually. The draft guidance states that as of 2004, “human activities annually produced more than 49 billion tons of GHG measured” in CO₂e.⁷ Thus, a source in the U.S. – or anywhere in the world – emitting 25,000

⁷ Intergovernmental Panel on Climate Change (IPCC), Fourth Assessment Report (2007).

metric tons of CO₂e annually represents only a mere 5/100,000th of 1 percent of the 49 billion metric tons of GHGs emitted globally. Even in the context of U.S. GHG emissions, which have been estimated to be about 7.2 billion tons annually, 25,000 tons would be less than 4/10,000th of 1 percent.

EPA's use of the 25,000-ton-per-year level is for a distinct purpose under entirely different legislation. According to the preamble of EPA's reporting rule, EPA adopted this level as a threshold for mandatory reporting by "owners and operators of certain fossil fuel facilities that directly emit" 25,000 metric tons of CO₂e or more per year, as determined by monitoring, reporting, and recordkeeping, as well as from "certain fossil fuel suppliers and industrial GHG suppliers" that are "located in the United States." See 40 C.F.R. §§ 98.1 and 3 as added by the rule.⁸

EPA chose a facility-level "approach because the purpose of the reporting rule is to collect data from suppliers and from facilities with direct GHG emissions above selected thresholds for use in analyzing, developing and implementing potential future . . . GHG policies and programs." None of these is a NEPA requirement, and none relates to the environmental impacts of the GHG emissions or the need to study such impacts.

⁸ On April 12, 2010, EPA proposed a "supplemental rule to require reporting" of GHGs from petroleum and natural gas suppliers and proposed the same threshold, although EPA "considered" but rejected lowering it to 10,000 metric tons of CO₂e. See 75 *Fed. Reg.* 18,608, 18,619.

Moreover, EPA recently issued the “tailoring rule,” which identifies initial levels of emissions to which certain PSD permitting provisions will apply. The GHG emissions thresholds will be 75,000 tons per year for modified sources and 100,000 tons per year for new sources between July 2011 and June 2013. See *Prevention of Significant Deterioration and Title V Greenhouse Gas Tailoring Rule*, Docket No. EPA-HQ-OAR-2009-0517 (signed May 13, 2010), pp. 52-53.

Furthermore, as discussed previously, GHG emissions are well-mixed globally and are ubiquitous. GHGs have no local impacts or “hot spots.” The global location of a project is irrelevant, whether in the U.S. or overseas. A qualitative NEPA analysis of a U.S. project would either ignore the GHG emissions from similar projects overseas or, as previously mentioned, could shift GHG emissions overseas. CEQ’s draft guidance does not properly reflect the global context of the GHG issue.

Given that annual emissions of 25,000 metric tons do not rise to the level of a discernible effect, we encourage CEQ to reconsider its recommendation for agencies to identify individual activity GHG emissions in undertaking NEPA analyses. If CEQ does retain such a recommendation, CEQ should emphasize that the purpose of identifying the emissions is simply to provide information on the nature of the proposed activity, not to point to any indicator of environmental effect.

D. CEQ Should Not Call for Speculative or Duplicative Analysis of Indirect Climate Effects.

CEQ’s NEPA regulations define “effects” to include both “direct” and “indirect” effects and, as to the latter, explain that they “are caused by the action and are later in time or farther removed

in distance, but are still reasonably foreseeable.” 40 C.F.R. § 1508.8(b). In the draft guidance, CEQ indicates that agencies must analyze indirect effects of climate change. CEQ states that it “does not propose” its 25,000-metric-ton “reference point for use as a measure of indirect effects” (p. 3). We agree with that decision. However, the Council does not indicate whether, how and under what circumstances agencies should consider indirect effects.

EEI encourages CEQ to clarify that agencies should not consider indirect effects that are speculative or unfounded. Thus, for example, if an agency is examining the environmental effects of a proposed electric transmission line intended to relieve general congestion on the grid, the agency should not be required to examine potential changes in GHG emissions from the wide array of generation facilities that may produce electricity that may be transported on that line. Transmission lines are single elements in a highly integrated grid that supports the delivery of electricity from generation sources to customers, and unless a line were specifically dedicated to serve particular new generating plants – a relatively rare occurrence given the interconnected nature of the grid – it would be purely speculative to guess what generation sources may produce electricity over a particular line now and into the future, much less to speculate on the GHG emissions that would be involved.

The extent to which a given generation facility is used to generate electricity at a given time of day depends more on fluctuations in demand for electricity driven by factors such as weather and fuel prices, than on the presence or absence of a single transmission line. Moreover, the mix of generation facilities evolves over time as a result of a complex array of factors, including air, water, coal ash and other environmental regulations. Similarly, the extent to which electricity

from different generation facilities may be transported over a given line is driven more by grid physics and bulk power contracts than the mere presence or absence of a single line. To require agencies to guess how a particular line might change GHG emissions would be to require them to engage in speculation that is inappropriate and would not provide meaningful information to assist them with their decisions.

In this regard, we note *Mayo Foundation v. Surface Transp. Bd.*, 472 F.3d 545 (8th Cir. 2006). Petitioner Sierra Club contended that the increased use of Powder River Basin (PRB) coal due to a new rail line would increase air-polluting emissions and that the Board did not adequately address this issue. The court had found in an earlier decision that the final EIS was “inadequate because it did not include information on the potential increase in emissions of other pollutants – specifically . . . carbon dioxide . . .” and noted that computer “programs could be used to forecast the effects of this project on the consumption of coal.” *Mid States Coalition for Progress v. Surface Transp. Bd.*, 345 F.3d 520, 548-50 (8th Cir. 2003).

On remand, the Board did examine potential impacts of the new line on air emissions and concluded that they were speculative and unpredictable. On subsequent review, the court noted that the Board had relied on EIA’s National Energy Modeling System (NEMS) after considering several other models because NEMS “not only forecasts coal supply and demand but also quantifies environmental impacts.” *Id.* at 555. Upholding the Board’s review, the court said:

After noting that “the impacts of this project on coal consumption and resulting air emissions would be small” on a national and regional basis and that any potential local air quality impacts were “speculative” and “ultimately unforeseeable,” the Board concluded that it was not necessary to impose additional mitigating conditions on the project.

Id at 555-56 (citations omitted). *See also Methow Valley Citizens Council v. Regional Forester*, 833 F.2d 810, 816 -817 (9th Cir.1987), *rev'd on other grounds*, *Robertson v. Methow Valley Citizens Council*, 490 U.S. 332 (1989) (the government is not required to speculate on impacts in order to “foresee the unforeseeable,” citing City of Davis v. Coleman, 521 F.2d 661, 676 (9th Cir.1975)).

EEI also encourages CEQ to clarify that agencies should not reconsider indirect effects that already have been analyzed or addressed in prior proceedings or are being analyzed or addressed in separate proceedings. Thus, for example, if GHG emissions from a given facility already are regulated or have been analyzed in permitting the facilities, agencies should not be required to reexamine those emissions as indirect effects of other facilities. CEQ’s regulations at 40 C.F.R. § 1502.20 encourage agencies to tier off of prior environmental analyses rather than redoing those analyses. Accordingly, CEQ should not require reconsideration, as indirect effects, of GHG emissions already taken into account directly in regulating or permitting the facilities producing those emissions.

III. Analysis Of Climate Change Effects on Proposed Major Federal Actions Also Is Highly Speculative And Should Necessarily Be Circumscribed.

The draft guidance proposes that federal agencies “should determine which climate change impacts warrant consideration in their EAs [environmental assessments] and EISs because of their impact on the analysis of the environmental effects of a proposed agency action” (p. 6). Citing the NEPA regulations (40 C.F.R. §§ 1500.4(g) and 1501.7) and CEQ’s 1981 Scoping Guidance process, the draft guidance states that agencies will “determine whether climate change

considerations warrant emphasis or de-emphasis” and when scoping the impact of climate change, “the sensitivity, location, and timeframe of a proposed action will determine the degree to which consideration of these predictions or projections is warranted.”⁹ *Id.* The draft guidance adds:

As with analysis of any other present or future environment or resource condition, the observed and projected effects of climate change that warrant consideration are most appropriately described as part of the current and future state of the proposed action’s “affected environment.” 40 CFR 1502.15. Based on that description of climate change effects that warrant consideration, the agency may assess the extent that the effects of the proposal for agency action or its alternatives will add to, modify, or mitigate those effects. Such effects may include, but are not limited to, effects on the environment, on public health and safety, and on vulnerable populations who are more likely to be adversely affected by climate change. The final analysis documents an agency assessment of the effects of the actions considered, including alternatives, on the affected environment.

Id.

⁹ It is unclear what the words “these predictions or projections” mean. Neither is defined nor explained in the draft guidance. However, Working Group I of the IPCC Summary for Policymakers distinguishes between the terms in its Annex I Glossary, saying that:

a projection is a potential future evolution of a quantity or set of quantities, often computed with the aid of a model. Projections are distinguished from *predictions* in order to emphasize that projections involved assumptions concerning, for example, future socio-economic technological developments that may or may not be realized, and are therefore subject to substantial *uncertainty*.”

The IPCC also defines and discusses the term “predictability” as follows:

Predictability. The extent to which future states of a system may be predicted based on knowledge of current and past states of the system. ... Since knowledge of the *climate system*’s past and current status is generally imperfect, as are the models that utilise this knowledge to produce a *climate prediction*, and since the climate system is inherently *nonlinear* and *chaotic*, predictability of the climate system is inherently limited. Even with arbitrarily accurate models and observations, there may still be limits to the predictability of such a nonlinear system.

pp. 138-39 (second emphasis added, italics in original and reference omitted).

As the draft guidance points out, such analysis would be most fruitful for projects “designed for long-term utility and located in areas that are considered vulnerable to specific effects of climate change (such as increasing sea level or ecological change) within the project’s lifetime.” *Id.* at 7. Accordingly, agencies should have considerable discretion to determine how they choose to plan for possible external impacts on their activities, including economic, social and environmental issues.

As to federal agencies applying climate change modeling in performing their NEPA analysis, the draft guidance suggests, citing Synthesis and Assessment Product (SAP) 3.1 on “Climate Models: An Assessment of Strengths and Limitations” and 40 C.F.R. §§ 1502.21-502.22, that such modeling is acceptable so long as the federal agencies “consider the uncertainties with long-term projections” from such global and regional models and “disclose . . . the extent to which they rely on particular studies or projections” because “[t]here are limitations and variability in the capacity” of those models “to reliably project potential changes at the regional, local, or project level” (p. 8) (emphasis added). This brief dissertation on the use and reliability of modeling ends by noting: “The outputs of coarse-resolution global climate models, commonly used to project climate change scenarios at a continental or regional scale, require downscaling and bias removal . . . before they can be used in regional or local impact studies.” *Id.* (emphasis added).

Accordingly, with all these caveats and no apparent resolution offered on how to overcome the shortcomings and “limitations,” CEQ first should assess whether use and reliance on such modeling by federal agencies would be practicable and not lead to claims of speculation.

Second, CEQ should dissuade agencies from applying these models unless and until they have the resources and expertise to overcome these limitations within an acceptable level of certainty.

With regard to GCM's, the aforementioned Global Climate Change Impacts in the United States report states, "All of the models used in this work have imperfections in their representation of the complexities of the 'real world' climate system . . . due to both limits in our understanding of the climate system, and in our ability to represent its complex behavior with available computer resources" (p. 22). Nevertheless, the report contends that "[d]espite this, models are extremely useful" and "are the only tools that exist for trying to understand the climate changes likely to be experienced over the course of this century." While such models may be useful in certain situations, the fact remains that they have limitations, particularly as they attempt to predict impacts regionally (*i.e.*, at the continental and sub-continental levels) and locally.

Indeed, the IPCC's Working Group I report, "The Science Basis," in the Third Assessment Report (2001) discussed some of those "limitations" in section 10.2.3 with respect to Regional Climate Models (RCMs), saying that this "modeling technique consists of initial conditions, time-dependent lateral meteorological conditions and surface boundary conditions to drive high-resolution RCMs," with the "driving data . . . derived from GCMs (or analyses of observations and can include GHG aerosol forcing." However, the section adds that "[t]he main limitations of this technique are the effects of systemic errors in the driving fields provided by global models; and lack of two-way interactions between regional and global climate" In addition, the "RCM simulations can be computationally demanding, which has limited the length of many experiments to date."

One potential approach to regional climate projection is “statistical downscaling,” a practice that does little to foster confidence. As discussed in the Working Group I report:

The major theoretical weakness of statistical downscaling methods is that their basic assumption is not verifiable, i.e., that the statistical relationships developed for present day climate also hold under the different forcing conditions of possible future climates. In addition, data with which to develop relationships may not be readily available in remote regions or regions with complex topography. Another caveat is that these empirically-based techniques cannot account for possible systematic changes in regional forcing conditions or feedback processes. The possibility of tailoring the statistical model to the requested regional or local information is a distinct advantage. However, it has the drawback that a systematic assessment of the uncertainty of this type of technique, as well as a comparison with other techniques, is difficult and may need to be carried out on a case-by-case basis.

p. 31, section 10.2.4 (emphasis added).

In the case of the IPCC’s Fourth Assessment Report, the Technical Summary also addresses

“Regional Downscaling.” Working Group I’s Summary for Policymakers states:

The availability of downscaling and other regionally focused studies remains uneven geographically, causing unevenness in the assessments that can be provided, particularly for extreme weather events. Downscaling studies demonstrate that local precipitation changes can vary significantly from those expected from the large-scale hydrological response pattern, particularly in areas of complex topography.

There remain a number of important sources of uncertainty limiting the ability to project regional climate change. While hydrological responses are relatively robust in certain core subpolar and subtropical regions, there is uncertainty in the precise location of these boundaries between increasing and decreasing precipitation. There are some important climate processes that have a significant effect on regional climate, but for which the climate change response is still poorly known. These include ENSO, the NAO, blocking, the thermohaline circulation and changes in tropical cyclone distribution. For those regions that have strong topographical controls on their climatic patterns, there is often insufficient climate change information at the fine spatial resolution of the topography. In some regions there has been only very limited research on extreme weather events. Further, the projected climate change signal becomes comparable to larger internal variability at smaller spatial and temporal scales, making it more difficult to utilise recent trends to evaluate model performance.

p. 74, Box TS.10 (references omitted and emphasis in original).

An April 8, 2004, article in Nature, titled “Modelers Deplore ‘Short-Termism’ On Climate,” reported on a meeting of regional climate modelers in Lind, Sweden: “Projections of climate change in say Florida or the Alps carry more political weight than vague warnings about global warming. And for almost two decades, the specialists in regional assessment have sought to make such projections,” but, according to the article, the modelers acknowledged that “their success has been limited.”

In the previously noted October 3, 2008, letter to U.S. fishery agencies, EPA also focused on the limitations:

The current tools for simulating climate change generally focus on global and regional-scale modeling. Global and regional-scale models lack the capability to represent explicitly many important small-scale processes. As a result, confidence in regional- and sub-regional-scale projections is lower than at the global scale. There is thus limited scientific capability in assessing, detecting, or measuring the relationship between emissions of GHGs such as CO₂ from a specific single source and any localized impact on a listed species, its habitat, or its members for purposes of ESA considerations.

p. 4 (emphasis added).

In addition, a May 2008 report of the Committee on Environment and Natural Resources of the National Science and Technology Council, titled “Scientific Assessment of the Effects of Global Change on the United States,” included a section on “Attribution of Observed Climate Change to Human Activities at the Global and Continental Scale.” That section “addresses the extent to

which observed climate changes at the global and continental or national scale can be attributed to global anthropogenic emissions of greenhouse gases.” It states:

Difficulties remain in attributing temperature changes at smaller than continental scales and over time scales of less than 50 years. Attribution results at these scales have, with limited exceptions, not been established. Averaging over smaller regions reduces the natural variability less than does averaging over large regions, making it more difficult to distinguish between changes expected from external forcing and variability. In addition, temperature changes associated with some modes of variability are poorly simulated by models in some regions and seasons. Furthermore, the small-scale details of external forcing and the response simulated by models are less credible than large-scale features (IPCC, 2007a).

p. 70.

Until high-resolution computer models can predict with some degree of certainty future impacts of GHG emissions, CEQ should not require federal agencies to engage in such speculative exercises. **While it is appropriate for the draft guidance to refer to uncertainty and the limitations of regional modeling, the requirements of “downscaling and bias removal” for GCMs, and the need for agencies, in considering utilizing such modeling in their NEPA analysis, to disclose limitations, we question, in light of the above discussion, why agencies should undergo the burden and expenditure of resources of even considering such regional or local applications.**

In short, in light of their uncertainties and limitations, use of GCMs for regional and local NEPA analyses would be too speculative and unproductive, probably for some time to come, and we question CEQ’s inclusion of them in the draft guidance.

IV. Conclusion

In summary, EEI encourages CEQ to revisit in some fundamental ways the proposed guidance for consideration of GHGs and climate change in the NEPA context. While we agree that it often will be possible for federal agencies to estimate the anticipated GHG emissions of their proposed activities, CEQ needs to ensure that agencies in no way equate individual project GHG emissions, at or above 25,000 tons per year, as a “significant effect” warranting preparation of an EIS.

When compared with nationwide or global GHG emissions, 25,000 tons per year is too low a threshold to be meaningful for purposes of NEPA analysis. Indeed, the most helpful thing CEQ can and should do in its guidance on GHGs and climate change is to indicate that individual project GHG emissions typically will be minuscule compared to global emissions and so do not need to be studied in any qualitative way or substantial detail in the NEPA context.

Given the lack of any way to measure a causal link between individual project GHG emissions and specific climate change impacts, CEQ should ensure that the guidance does not require NEPA analyses to discuss the effect of an activity’s emissions on climate change, either quantitatively or qualitatively. Instead, the guidance should be limited to requiring publication of the activity’s projected annual emissions levels.

Finally, we question the ability of federal agencies to examine in a meaningful way the impact of climate change on a proposed activity – given that climate change modeling is fraught with

uncertainty, in particular when trying to ascertain climate change at a regional or local level – or with any degree of accuracy into the future. At most, agencies should be encouraged to

1) periodically check the evolution of climate change models and assess whether, in the future, these models have become sufficiently advanced so as to produce sound and robust regional and local estimates of specific environmental effects that may be useful to federal decision-makers, and 2) perhaps now only undertake a very simple and modest “scenario” analysis of potential impacts of climate change on their activity, to the extent that can be hypothesized, to see what aspects of the activity may need to be monitored as climate science evolves.

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**EDISON ELECTRIC
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QUINLAN J. SHEA, III
Executive Director, Environment

May 24, 2010

Ted Boling, Esq.
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Council on Environmental Quality
722 Jackson Place, NW
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Submitted by e-mail to: gcc.guidance@ceq.eop.gov

Re: Draft CEQ NEPA Guidance on Consideration of Climate Change
and Greenhouse Gases, 75 Fed. Reg. 8046 (February 23, 2010)

Dear Mr. Boling:

The Edison Electric Institute is pleased to submit the attached comments to the Council on Environmental Quality on the above-referenced “Draft NEPA Guidance on Consideration of the Effects of Climate Change and Greenhouse Gas Emissions,” dated February 18, 2010.

EEI appreciates the opportunity to submit these comments. If you have any questions about them, or need additional information, please contact me, Edward Comer at (ecomer@eei.org; 202-508-5615), William Fang at (bfang@eei.org; 202-508-5617) or Henri Bartholomot at (hbartholomot@eei.org; 202- 508-5622).

Respectfully submitted,

A handwritten signature in black ink, appearing to read 'Quinlan J. Shea', with a stylized flourish at the end.

Quinlan Shea
Executive Director, Environment

QS:cfb
Attachment

**COMMENTS OF THE EDISON ELECTRIC INSTITUTE
ON THE COUNCIL ON ENVIRONMENTAL QUALITY'S
"DRAFT NEPA GUIDANCE ON CONSIDERATION OF THE EFFECTS OF
CLIMATE CHANGE AND GREENHOUSE GAS EMISSIONS"**

May 24, 2010

The Edison Electric Institute (EEI) is pleased to submit these comments to the Council on Environmental Quality (CEQ or Council) on its "Draft NEPA Guidance on Consideration of the Effects of Climate Change and Greenhouse Gas Emissions," dated February 18, 2010. CEQ announced that the guidance was available, and invited comments by May 24, 2010, in a notice published at 75 *Fed. Reg.* 8046 (Feb. 23, 2010).

In that notice, CEQ specifically invited comments on "when and how Federal agencies must consider the impacts of proposed Federal actions on global climate change, as well as the expected environmental effects from climate change that may be relevant to the design of the proposed Federal action." EEI addresses both of these issues in our comments.

EEI is the association of shareholder-owned electric utilities in the U.S. Our members serve 95 percent of the ultimate customers in the shareholder-owned segment of the industry, and they represent approximately 70 percent of the nation's electric power industry. Much of the electricity required for homes and businesses is generated through means that produce carbon dioxide (CO₂), and utility operations may result in emissions of other greenhouse gases (GHGs). In addition, EEI members are affected directly as permittees, and indirectly through their reliance on federal agency resource decisions, by federal agency implementation of the National

Environmental Policy Act of 1969, 42 U.S.C. § 4321 *et seq.* (NEPA). Therefore, EEI has a direct interest in the draft guidance.

I. Overview

EEI is submitting these comments to CEQ because the power sector has a major interest in ensuring that environmental reviews of electric generation, transmission, and distribution projects and other such critical infrastructure – including fuel resources – are appropriate in nature, scope and content. We also seek to ensure that the environmental review process for such facilities is as streamlined and efficient as possible in order to avoid unnecessary delays and increased costs to power sector customers in obtaining necessary agency decisions as to the facilities.

EEI supports comprehensive, economy-wide climate change legislation. Comprehensive legislation is simply a better fit for addressing climate change than traditional environmental regulation under statutes such as the Clean Air Act (CAA), Clean Water Act, Endangered Species Act (ESA) and NEPA. Well-designed legislation can achieve significant GHG emissions reductions – for example, by promoting enhanced energy efficiency, lower GHG-emitting technology, renewable energy sources, advanced coal technologies coupled with carbon capture and storage (CCS), and other critical technologies and measures – while helping to maintain both the reliability and affordability of the nation’s electric energy supply.

We fully support the intent of NEPA to ensure that federal agencies consider significant environmental effects of their decisions on the human environment. At the same time, the NEPA

review process and decisions based on it need to be reasonable and timely in order to ensure the best use of limited federal agency, regulated community and public resources.

Agency reviews under NEPA already provide a comprehensive and sound review of proposed federal actions and their environmental impacts. Therefore, in order to avoid confusion, duplication, litigation and delay, CEQ guidance should be properly tailored to help to ensure consistency across federal agencies and reasonable, streamlined implementation of NEPA. To achieve these goals, guidance for considering climate change and GHGs in the NEPA context must be crafted with great care. GHG emissions are unlike traditional pollutants regulated under various environmental statutes because there is little (if any) relationship between GHG emissions from a project in a particular location and the possible environmental effects of climate change in that location. Instead, it is the total global accumulation of GHG emissions over a long period of time that matters. In addition, future climate conditions cannot be predicted or quantified with certainty at a global level, much less at a national, regional or local level. Furthermore, given the current state of general circulation models (GCMs), the science cannot definitively predict climate change effects resulting from GHG emissions at individual facilities at any level – national, regional or local. This makes the process of considering the effects of GHGs and climate change from individual project proposals inherently speculative. While CEQ has recognized this in part in the draft guidance, the final guidance should more fully reflect this basic reality in order to avoid encouraging federal agencies to spend time and funds on GHG and climate analyses that, in the end, would not meaningfully inform the agency decisions or even worse, could provide confusing or misleading information.

In addition, CEQ's guidance should reflect more clearly that CEQ's proposed reference level of 25,000 metric tons per year of CO₂ equivalent (CO₂e) does not constitute a basis for requiring more detailed environmental analysis of a particular activity. As we explain below, the 25,000-ton indicator was developed to collect emissions data, not as a regulatory benchmark. In fact, 25,000 tons represents only 5/100,000th of 1 percent (0.0000005) of the 49 billion tons of global GHG emissions in 2004. EPA's so-called "tailoring" rule, issued on May 13, 2010, will apply 75,000 and 100,000 tons per year as the minimum levels for triggering Prevention of Significant Deterioration (PSD) permits under the CAA. While it may be appropriate to report projected GHG emissions at such levels in an environmental impact statement (EIS) in order to inform the decision-maker and the public, even a 100,000-tons-per-year project will have an imperceptible impact upon climate, and therefore agencies should not be tasked with engaging in any further analysis that would be largely speculation.

While EEI appreciates CEQ's desire to provide guidance on how to consider climate change and GHGs in NEPA reviews, we encourage the Council to refine the proposed guidance to provide greater clarity and more reasonable recommendations on how to accomplish this. At most, the guidance should require agencies to include information about anticipated GHG emissions in environmental analyses of major projects. The guidance should not require agencies to engage in qualitative discussion of the impacts of such emissions upon climate because any such changes are far too speculative to predict. Nor should agencies be required to conduct detailed environmental analyses, including EISs, simply because of an activity's GHG emissions. Instead, publication of projected annual emissions levels should be sufficient under NEPA analysis.

II. Because There Is No Substantive Basis To Draw A Link Between A Particular Project's GHG Emissions And Climate Change, Discussion Of The Effects Of Such Emissions On Climate Change And Alternatives For Reducing Those Emissions Would Be Speculative And Unproductive.

In the draft guidance, CEQ correctly concedes that “it is not currently useful” to attempt to make a causal link between climate change, or the environmental impacts of such change, and a particular project or its emissions because “such direct linkage is difficult to isolate and to understand” (p. 3). CEQ also acknowledges the view of the Environmental Protection Agency (EPA) that it is not currently “possible to quantify with great specificity (i.e. geographic) the various health effects from climate change.”¹ *Id.* at 11. In addition, CEQ recognizes that the Supreme Court has stated that the obligation of an agency to discuss particular effects turns on “a reasonably close causal relationship between the environmental effect and the alleged cause.” *Id.* at 7, quoting *Dep’t of Transp. v. Public Citizen*, 541 U.S. 752, 767 (2004)).

Because there is no such demonstrable “causal relationship” between an individual activity’s GHG emissions and specific climate change impacts, there is no meaningful basis for requiring agencies to engage in a speculative discussion of such a link in the NEPA context. Yet the draft

¹ In the proceeding where EPA made endangerment findings under the CAA, the EPA Administrator said:

The Administrator recognizes that human-induced climate change has the potential to be far-reaching and multi-dimensional, and in light of existing knowledge, that not all risks and potential impacts can be quantified or characterized with uniform metrics. There is variety not only in the nature and potential magnitude of risks and impacts, but also in our ability to characterize, quantify and project such impacts into the future.

74 *Fed. Reg.* 66,496, 66,497 (2009) (emphasis added).

guidance says that it would be appropriate, in a federal agency's analysis of an activity's direct effects under NEPA, for the agency to quantify the activity's GHG emissions, to discuss "qualitatively" the link between those emissions and climate change, and to consider measures to reduce the emissions, including consideration of reasonable alternatives. *Id.* at 3.

While EEI agrees that agencies can provide some reasonable estimates of GHG emissions anticipated from individual activities, we strongly disagree that agencies can or should be expected to discuss in their NEPA analyses the impact of an activity's GHG emissions on climate change, or the impacts of climate change on the activity, because such discussions would be speculative and thus unproductive and unhelpful to agency decision-making.

A. Courts Agree that under NEPA, Agencies Need Not Analyze Effects Absent a Causal Link and Need Not Engage in Speculative Alternatives Analysis.

The courts have long applied a "rule of reason" to NEPA analysis, which holds that agencies are not required to engage in speculation or "crystal ball inquiry." In *National Resources Defense Council, Inc. v. Morton*, 458 F.2d 827 (D.C. Cir. 1972), the court examined the need to consider reasonable alternatives in an EIS in the context of an oil and gas lease sale under the Outer Continental Shelf Lands Act and concluded:

[T]he requirement in NEPA of discussion as to reasonable alternatives does not require "crystal ball" inquiry. ... The statute must be construed in the light of reason if it is not to demand what is, fairly speaking, not meaningfully possible, given the obvious, that the resources of energy and research — and time — available to meet the Nation's needs are not infinite. ... In the last analysis, the requirement as to alternatives is subject to a construction of reasonableness, and we say this with full awareness that this approach necessarily has both strengths and weaknesses.

458 F.2d at 837-38 (emphasis added). CEQ's current regulations adopt these same principles.

40 C.F.R. §§ 1500.4(f)-(g), 1501.7 and 1508.25. The draft guidance states that "[a]gencies apply

the rule of reason” on the usefulness of information regarding the “proposal, its alternatives, and mitigations options” (p. 5). It also appropriately refers to *Public Citizen*, supra, where the Supreme Court held:

[I]nherent in NEPA and its implementing regulations is a “rule of reason,” which ensures that agencies determine whether and to what extent to prepare an EIS based on the usefulness of any new potential information to the decisionmaking process. See *Marsh*, 490 U.S., at 373-374, 109 S.Ct. 1851. Where the preparation of an EIS would serve “no purpose” in light of NEPA’s regulatory scheme as a whole, no rule of reason worthy of that title would require an agency to prepare an EIS. See *Aberdeen & Rockfish R. Co. v. Students Challenging Regulatory Agency Procedures (SCRAP)*, 422 U.S. 289, 325, 95 S.Ct. 2336, 45 L.Ed.2d 191 (1975); see also 40 CFR §§ 1500.1(b)-(c) (2003).

541 U.S. at 767-68.

Federal agencies should not be encouraged to engage in “qualitative” analysis of the effects of GHG emissions or alternatives for reducing GHGs if that would require a “crystal ball inquiry.” The “rule of reason” must apply when effects from an individual project proposal cannot be readily ascertained and the value (if any) of the alternatives is remote and speculative.

B. There Is No Demonstrable Causal Link between an Activity’s GHG Emissions and Climate Change, and Alternatives Analysis in this Context is Inherently Speculative.

The draft guidance accurately states that “it is not currently useful for the NEPA analysis to attempt to link specific climatological changes, or the environmental impacts thereof, to the particular project or emissions, as such direct linkage is difficult to isolate and understand” (p. 3). This position is consistent with the positions taken by other federal agencies.

In a May 14, 2008, memorandum, the Department of the Interior (DOI) stated, “It is currently beyond the scope of existing science to identify a specific source of CO₂ emissions and designate

it as the cause of specific climate impacts at an exact location.” Mark D. Myers, Director, U.S.

Geological Survey re: “The Challenges of Linking Carbon Emissions, Atmospheric Greenhouse Gas Concentrations, Global Warming, and Consequential Impacts” (emphasis added).

Subsequently, in an October 3, 2008, letter to the Director of the U.S. Fish and Wildlife Service and the Director of the Office of Protected Resources, National Marine Fisheries Service regarding the “Endangered Species Act and GHG Emitting Activities,” EPA cited the DOI memorandum and agreed that current scientific analysis lacked the “tools” to make the “linkage” referred to by CEQ. EPA’s Principal Deputy Assistant Administrator, Office of Air and Radiation stated:

To date, research on how emissions of CO₂ and other GHGs influence global climate change and associated effects has focused on the overall impact of emissions from aggregate regional or global sources. This is primarily because GHG emissions from single sources are small relative to aggregate emissions, and GHGs, once emitted from a given source, become well mixed in the global atmosphere and have a long atmospheric lifetime. The climate change research community has not yet developed tools specifically intended for evaluating or quantifying end-point impacts attributable to the emissions of GHGs from a single source, and we are not aware of any scientific literature to draw from regarding the climate effects of individual, facility-level GHG emissions.

p. 4 (emphasis added).

In short, until the science of GCMs improves, no one will be able to model with any certainty or reliably predict the effects of a particular project’s GHG emissions on climate change. In recognition of the fact that such modeling would be an exercise in sheer speculation, the Obama Administration is initiating an almost \$50-million, five-year research effort by the National Science Foundation and the Agriculture and Energy Departments “aimed at producing new, high-resolution computer models” with the goals of producing “reliable predictions of climate

change at regional decadal scales, the resulting impacts, and the potential adaptation of living systems to these impacts.”²

Climate change is a global phenomenon caused by cumulative global GHG emissions over long periods of time, and is not a national, regional or local phenomenon linked to individual project emissions.³ Thus, the draft guidance calls climate change a “global problem that results from global GHG emissions,” while appropriately noting that “there are more sources and actions emitting GHGs (in terms of both absolute numbers and types) than are typically encountered when evaluating the emissions of other pollutants” (p. 2). Indeed, the Economic Report of the President to Congress (February 2010) observes, “Developing countries such as China and India are responsible for a growing proportion of emissions because of their heavy reliance on carbon-intensive fuels, such as coal . . .” (p. 255).

Both the Administration and Congress have recognized that addressing climate change will require concerted action globally, through international treaties and national actions, including enactment of economy-wide congressional legislation in the U.S., that together address the matter comprehensively and effectively. Furthermore, there are no GHG “hot spots,” such as occurs with some traditional air pollutants. GHG emissions do not have localized effects. For

² Energy Daily (March 25, 2010).

³ The Energy Information Administration’s (EIA) International Energy Outlook 2009, Chapter 8, titled “Energy-Related Carbon Dioxide Emissions,” states, “Because anthropogenic emissions of carbon dioxide result primarily from the combustion of fossil fuels, world energy use continues to be at the center of the climate debate” (p. 109).

these reasons, EPA has not targeted GHG emissions for regulation under the CAA in any regulation aimed at improving air quality in specific localities, states or regions of the U.S.

Quantitatively, the draft guidance states that this “problem is much more the result of numerous and varied sources, each of which might seem to make a relatively small addition to global atmospheric GHG concentrations” (p. 2). While the draft guidance briefly recognizes the global context of GHG emissions and climate change, CEQ appears to be attempting to mitigate the impacts of climate change at a local project level under a statute that is ill-suited to address the global nature of GHG emissions. This is precisely why Congress is attempting to legislate an appropriate response to the issue rather than leaving it to EPA under the CAA to regulate GHG emissions in inappropriate, costly and awkward ways.

Furthermore, the nature of climate change raises the problem of “leakage.” If NEPA analysis of GHG emissions and climate change should stall or stop any specific project, this could in many cases simply shift the emissions overseas, perhaps even increasing net global emissions.⁴ The energy-related CO₂ emissions of non-Organization of Economic Cooperation and Development (OECD) nations surpassed the CO₂ emissions of OECD countries in 2006, and by 2030 non-

⁴ In its April 28, 2010, report to Congress titled EPA Regulation of Greenhouse Gases: Congressional Responses and Options, the Congressional Research Service concluded:

The differences are of scale and of degree. Greenhouse gases are global pollutants to a greater extent than most of the pollutants previously regulated under the [Clean Air] Act; reductions in U.S. emissions without simultaneous reductions by other countries may somewhat diminish but will not solve the problems the emissions cause.

pp. 12-13 (footnote omitted and emphasis added).

OECD emissions are projected to be 73 percent greater than those of OECD nations.⁵ Since the already minuscule contribution of any specific federal project's GHG emissions to cumulative global emissions will only become smaller and smaller in the future, it would counterproductive to shift those emissions overseas.

C. 25,000 Metric Tons Per Year Should Not Be a Trigger for Quantitative Analysis.

The draft guidance chooses as a trigger 25,000 metric tons or more of CO₂e GHG emissions and indicates that “if a proposed action would be reasonably anticipated to cause direct emissions” in that quantity annually, “agencies should consider this as an indicator that a quantitative and qualitative assessment may be meaningful to decision makers and the public” (p. 1) (emphasis added). The draft guidance adds that in the case of “long-term actions that have annual direct emissions of less than” such quantity, CEQ “encourages” the agencies “to consider whether the action’s long-term emissions should receive similar analysis.” *Id.* at 1-2.

The draft guidance then states that CEQ is “not” proposing this metric “as an indicator of a threshold of significant effects, but rather as an indicator of a minimum level” of emissions “that may warrant some description in the appropriate NEPA analysis” of agency actions “involving direct” GHG emissions. *Id.* at 2. The draft guidance also notes that this number now “serves as a minimum standard for reporting emissions” that has been recently promulgated by EPA.⁶ *Id.* at 3. The draft guidance contends that the 25,000-metric-tons figure “may provide a useful, presumptive, threshold for discussion and disclosure of GHG emissions because it has been used

⁵ See EIA, *supra* n. 3, at 6 (Fig. 9), 109 (Fig. 80).

⁶ We disagree with CEQ’s characterization of the EPA reporting rules as being issued under the CAA. In fact the initial reporting rule was issued under the Omnibus Appropriations Act of FY08.

and proposed” by EPA and “because it provides comprehensive coverage of emissions . . . , thereby creating an important data set useful in quantitative analyses of GHG policies, programs and regulations” and, as such, this “rationale is pertinent to the presentation of NEPA analysis as well.” *Id.* at 3 n. 2.

While it is reasonable for CEQ to attempt to provide a benchmark level above which agencies should examine GHG emissions from a proposed activity in undertaking NEPA analyses, we are concerned that – notwithstanding CEQ comments in the draft guidance to the contrary – agencies would improperly treat this low level of emissions as a “significant effect” threshold for NEPA purposes.

First, CEQ has failed to provide any scientific justification for the 25,000-ton indicator. If science is truly to be the foundation for the Administration’s environmental policy, there should be a principled rationale for such a number.

Second, an individual project’s GHG emissions – even for projects with relatively large emissions – will be infinitesimal compared to worldwide GHG emissions, and more importantly the individual project’s emissions will not have a demonstrable effect on climate change. This is clearly true for projects emitting 25,000 tons of GHG emissions annually. The draft guidance states that as of 2004, “human activities annually produced more than 49 billion tons of GHG measured” in CO₂e.⁷ Thus, a source in the U.S. – or anywhere in the world – emitting 25,000

⁷ Intergovernmental Panel on Climate Change (IPCC), Fourth Assessment Report (2007).

metric tons of CO₂e annually represents only a mere 5/100,000th of 1 percent of the 49 billion metric tons of GHGs emitted globally. Even in the context of U.S. GHG emissions, which have been estimated to be about 7.2 billion tons annually, 25,000 tons would be less than 4/10,000th of 1 percent.

EPA's use of the 25,000-ton-per-year level is for a distinct purpose under entirely different legislation. According to the preamble of EPA's reporting rule, EPA adopted this level as a threshold for mandatory reporting by "owners and operators of certain fossil fuel facilities that directly emit" 25,000 metric tons of CO₂e or more per year, as determined by monitoring, reporting, and recordkeeping, as well as from "certain fossil fuel suppliers and industrial GHG suppliers" that are "located in the United States." See 40 C.F.R. §§ 98.1 and 3 as added by the rule.⁸

EPA chose a facility-level "approach because the purpose of the reporting rule is to collect data from suppliers and from facilities with direct GHG emissions above selected thresholds for use in analyzing, developing and implementing potential future . . . GHG policies and programs." None of these is a NEPA requirement, and none relates to the environmental impacts of the GHG emissions or the need to study such impacts.

⁸ On April 12, 2010, EPA proposed a "supplemental rule to require reporting" of GHGs from petroleum and natural gas suppliers and proposed the same threshold, although EPA "considered" but rejected lowering it to 10,000 metric tons of CO₂e. See 75 *Fed. Reg.* 18,608, 18,619.

Moreover, EPA recently issued the “tailoring rule,” which identifies initial levels of emissions to which certain PSD permitting provisions will apply. The GHG emissions thresholds will be 75,000 tons per year for modified sources and 100,000 tons per year for new sources between July 2011 and June 2013. See *Prevention of Significant Deterioration and Title V Greenhouse Gas Tailoring Rule*, Docket No. EPA-HQ-OAR-2009-0517 (signed May 13, 2010), pp. 52-53.

Furthermore, as discussed previously, GHG emissions are well-mixed globally and are ubiquitous. GHGs have no local impacts or “hot spots.” The global location of a project is irrelevant, whether in the U.S. or overseas. A qualitative NEPA analysis of a U.S. project would either ignore the GHG emissions from similar projects overseas or, as previously mentioned, could shift GHG emissions overseas. CEQ’s draft guidance does not properly reflect the global context of the GHG issue.

Given that annual emissions of 25,000 metric tons do not rise to the level of a discernible effect, we encourage CEQ to reconsider its recommendation for agencies to identify individual activity GHG emissions in undertaking NEPA analyses. If CEQ does retain such a recommendation, CEQ should emphasize that the purpose of identifying the emissions is simply to provide information on the nature of the proposed activity, not to point to any indicator of environmental effect.

D. CEQ Should Not Call for Speculative or Duplicative Analysis of Indirect Climate Effects.

CEQ’s NEPA regulations define “effects” to include both “direct” and “indirect” effects and, as to the latter, explain that they “are caused by the action and are later in time or farther removed

in distance, but are still reasonably foreseeable.” 40 C.F.R. § 1508.8(b). In the draft guidance, CEQ indicates that agencies must analyze indirect effects of climate change. CEQ states that it “does not propose” its 25,000-metric-ton “reference point for use as a measure of indirect effects” (p. 3). We agree with that decision. However, the Council does not indicate whether, how and under what circumstances agencies should consider indirect effects.

EEI encourages CEQ to clarify that agencies should not consider indirect effects that are speculative or unfounded. Thus, for example, if an agency is examining the environmental effects of a proposed electric transmission line intended to relieve general congestion on the grid, the agency should not be required to examine potential changes in GHG emissions from the wide array of generation facilities that may produce electricity that may be transported on that line. Transmission lines are single elements in a highly integrated grid that supports the delivery of electricity from generation sources to customers, and unless a line were specifically dedicated to serve particular new generating plants – a relatively rare occurrence given the interconnected nature of the grid – it would be purely speculative to guess what generation sources may produce electricity over a particular line now and into the future, much less to speculate on the GHG emissions that would be involved.

The extent to which a given generation facility is used to generate electricity at a given time of day depends more on fluctuations in demand for electricity driven by factors such as weather and fuel prices, than on the presence or absence of a single transmission line. Moreover, the mix of generation facilities evolves over time as a result of a complex array of factors, including air, water, coal ash and other environmental regulations. Similarly, the extent to which electricity

from different generation facilities may be transported over a given line is driven more by grid physics and bulk power contracts than the mere presence or absence of a single line. To require agencies to guess how a particular line might change GHG emissions would be to require them to engage in speculation that is inappropriate and would not provide meaningful information to assist them with their decisions.

In this regard, we note *Mayo Foundation v. Surface Transp. Bd.*, 472 F.3d 545 (8th Cir. 2006). Petitioner Sierra Club contended that the increased use of Powder River Basin (PRB) coal due to a new rail line would increase air-polluting emissions and that the Board did not adequately address this issue. The court had found in an earlier decision that the final EIS was “inadequate because it did not include information on the potential increase in emissions of other pollutants – specifically . . . carbon dioxide . . .” and noted that computer “programs could be used to forecast the effects of this project on the consumption of coal.” *Mid States Coalition for Progress v. Surface Transp. Bd.*, 345 F.3d 520, 548-50 (8th Cir. 2003).

On remand, the Board did examine potential impacts of the new line on air emissions and concluded that they were speculative and unpredictable. On subsequent review, the court noted that the Board had relied on EIA’s National Energy Modeling System (NEMS) after considering several other models because NEMS “not only forecasts coal supply and demand but also quantifies environmental impacts.” *Id.* at 555. Upholding the Board’s review, the court said:

After noting that “the impacts of this project on coal consumption and resulting air emissions would be small” on a national and regional basis and that any potential local air quality impacts were “speculative” and “ultimately unforeseeable,” the Board concluded that it was not necessary to impose additional mitigating conditions on the project.

Id at 555-56 (citations omitted). *See also Methow Valley Citizens Council v. Regional Forester*, 833 F.2d 810, 816 -817 (9th Cir.1987), *rev'd on other grounds, Robertson v. Methow Valley Citizens Council*, 490 U.S. 332 (1989) (the government is not required to speculate on impacts in order to “foresee the unforeseeable,” citing City of Davis v. Coleman, 521 F.2d 661, 676 (9th Cir.1975)).

EEI also encourages CEQ to clarify that agencies should not reconsider indirect effects that already have been analyzed or addressed in prior proceedings or are being analyzed or addressed in separate proceedings. Thus, for example, if GHG emissions from a given facility already are regulated or have been analyzed in permitting the facilities, agencies should not be required to reexamine those emissions as indirect effects of other facilities. CEQ’s regulations at 40 C.F.R. § 1502.20 encourage agencies to tier off of prior environmental analyses rather than redoing those analyses. Accordingly, CEQ should not require reconsideration, as indirect effects, of GHG emissions already taken into account directly in regulating or permitting the facilities producing those emissions.

III. Analysis Of Climate Change Effects on Proposed Major Federal Actions Also Is Highly Speculative And Should Necessarily Be Circumscribed.

The draft guidance proposes that federal agencies “should determine which climate change impacts warrant consideration in their EAs [environmental assessments] and EISs because of their impact on the analysis of the environmental effects of a proposed agency action” (p. 6). Citing the NEPA regulations (40 C.F.R. §§ 1500.4(g) and 1501.7) and CEQ’s 1981 Scoping Guidance process, the draft guidance states that agencies will “determine whether climate change

considerations warrant emphasis or de-emphasis” and when scoping the impact of climate change, “the sensitivity, location, and timeframe of a proposed action will determine the degree to which consideration of these predictions or projections is warranted.”⁹ *Id.* The draft guidance adds:

As with analysis of any other present or future environment or resource condition, the observed and projected effects of climate change that warrant consideration are most appropriately described as part of the current and future state of the proposed action’s “affected environment.” 40 CFR 1502.15. Based on that description of climate change effects that warrant consideration, the agency may assess the extent that the effects of the proposal for agency action or its alternatives will add to, modify, or mitigate those effects. Such effects may include, but are not limited to, effects on the environment, on public health and safety, and on vulnerable populations who are more likely to be adversely affected by climate change. The final analysis documents an agency assessment of the effects of the actions considered, including alternatives, on the affected environment.

Id.

⁹ It is unclear what the words “these predictions or projections” mean. Neither is defined nor explained in the draft guidance. However, Working Group I of the IPCC Summary for Policymakers distinguishes between the terms in its Annex I Glossary, saying that:

a projection is a potential future evolution of a quantity or set of quantities, often computed with the aid of a model. Projections are distinguished from *predictions* in order to emphasize that projections involved assumptions concerning, for example, future socio-economic technological developments that may or may not be realized, and are therefore subject to substantial *uncertainty*.”

The IPCC also defines and discusses the term “predictability” as follows:

Predictability. The extent to which future states of a system may be predicted based on knowledge of current and past states of the system. ... Since knowledge of the *climate system*’s past and current status is generally imperfect, as are the models that utilise this knowledge to produce a *climate prediction*, and since the climate system is inherently *nonlinear* and *chaotic*, predictability of the climate system is inherently limited. Even with arbitrarily accurate models and observations, there may still be limits to the predictability of such a nonlinear system.

pp. 138-39 (second emphasis added, italics in original and reference omitted).

As the draft guidance points out, such analysis would be most fruitful for projects “designed for long-term utility and located in areas that are considered vulnerable to specific effects of climate change (such as increasing sea level or ecological change) within the project’s lifetime.” *Id.* at 7. Accordingly, agencies should have considerable discretion to determine how they choose to plan for possible external impacts on their activities, including economic, social and environmental issues.

As to federal agencies applying climate change modeling in performing their NEPA analysis, the draft guidance suggests, citing Synthesis and Assessment Product (SAP) 3.1 on “Climate Models: An Assessment of Strengths and Limitations” and 40 C.F.R. §§ 1502.21-502.22, that such modeling is acceptable so long as the federal agencies “consider the uncertainties with long-term projections” from such global and regional models and “disclose . . . the extent to which they rely on particular studies or projections” because “[t]here are limitations and variability in the capacity” of those models “to reliably project potential changes at the regional, local, or project level” (p. 8) (emphasis added). This brief dissertation on the use and reliability of modeling ends by noting: “The outputs of coarse-resolution global climate models, commonly used to project climate change scenarios at a continental or regional scale, require downscaling and bias removal . . . before they can be used in regional or local impact studies.” *Id.* (emphasis added).

Accordingly, with all these caveats and no apparent resolution offered on how to overcome the shortcomings and “limitations,” CEQ first should assess whether use and reliance on such modeling by federal agencies would be practicable and not lead to claims of speculation.

Second, CEQ should dissuade agencies from applying these models unless and until they have the resources and expertise to overcome these limitations within an acceptable level of certainty.

With regard to GCM's, the aforementioned Global Climate Change Impacts in the United States report states, "All of the models used in this work have imperfections in their representation of the complexities of the 'real world' climate system . . . due to both limits in our understanding of the climate system, and in our ability to represent its complex behavior with available computer resources" (p. 22). Nevertheless, the report contends that "[d]espite this, models are extremely useful" and "are the only tools that exist for trying to understand the climate changes likely to be experienced over the course of this century." While such models may be useful in certain situations, the fact remains that they have limitations, particularly as they attempt to predict impacts regionally (*i.e.*, at the continental and sub-continental levels) and locally.

Indeed, the IPCC's Working Group I report, "The Science Basis," in the Third Assessment Report (2001) discussed some of those "limitations" in section 10.2.3 with respect to Regional Climate Models (RCMs), saying that this "modeling technique consists of initial conditions, time-dependent lateral meteorological conditions and surface boundary conditions to drive high-resolution RCMs," with the "driving data . . . derived from GCMs (or analyses of observations and can include GHG aerosol forcing." However, the section adds that "[t]he main limitations of this technique are the effects of systemic errors in the driving fields provided by global models; and lack of two-way interactions between regional and global climate" In addition, the "RCM simulations can be computationally demanding, which has limited the length of many experiments to date."

One potential approach to regional climate projection is “statistical downscaling,” a practice that does little to foster confidence. As discussed in the Working Group I report:

The major theoretical weakness of statistical downscaling methods is that their basic assumption is not verifiable, i.e., that the statistical relationships developed for present day climate also hold under the different forcing conditions of possible future climates. In addition, data with which to develop relationships may not be readily available in remote regions or regions with complex topography. Another caveat is that these empirically-based techniques cannot account for possible systematic changes in regional forcing conditions or feedback processes. The possibility of tailoring the statistical model to the requested regional or local information is a distinct advantage. However, it has the drawback that a systematic assessment of the uncertainty of this type of technique, as well as a comparison with other techniques, is difficult and may need to be carried out on a case-by-case basis.

p. 31, section 10.2.4 (emphasis added).

In the case of the IPCC’s Fourth Assessment Report, the Technical Summary also addresses

“Regional Downscaling.” Working Group I’s Summary for Policymakers states:

The availability of downscaling and other regionally focused studies remains uneven geographically, causing unevenness in the assessments that can be provided, particularly for extreme weather events. Downscaling studies demonstrate that local precipitation changes can vary significantly from those expected from the large-scale hydrological response pattern, particularly in areas of complex topography.

There remain a number of important sources of uncertainty limiting the ability to project regional climate change. While hydrological responses are relatively robust in certain core subpolar and subtropical regions, there is uncertainty in the precise location of these boundaries between increasing and decreasing precipitation. There are some important climate processes that have a significant effect on regional climate, but for which the climate change response is still poorly known. These include ENSO, the NAO, blocking, the thermohaline circulation and changes in tropical cyclone distribution. For those regions that have strong topographical controls on their climatic patterns, there is often insufficient climate change information at the fine spatial resolution of the topography. In some regions there has been only very limited research on extreme weather events. Further, the projected climate change signal becomes comparable to larger internal variability at smaller spatial and temporal scales, making it more difficult to utilise recent trends to evaluate model performance.

p. 74, Box TS.10 (references omitted and emphasis in original).

An April 8, 2004, article in Nature, titled “Modelers Deplore ‘Short-Termism’ On Climate,” reported on a meeting of regional climate modelers in Lind, Sweden: “Projections of climate change in say Florida or the Alps carry more political weight than vague warnings about global warming. And for almost two decades, the specialists in regional assessment have sought to make such projections,” but, according to the article, the modelers acknowledged that “their success has been limited.”

In the previously noted October 3, 2008, letter to U.S. fishery agencies, EPA also focused on the limitations:

The current tools for simulating climate change generally focus on global and regional-scale modeling. Global and regional-scale models lack the capability to represent explicitly many important small-scale processes. As a result, confidence in regional- and sub-regional-scale projections is lower than at the global scale. There is thus limited scientific capability in assessing, detecting, or measuring the relationship between emissions of GHGs such as CO₂ from a specific single source and any localized impact on a listed species, its habitat, or its members for purposes of ESA considerations.

p. 4 (emphasis added).

In addition, a May 2008 report of the Committee on Environment and Natural Resources of the National Science and Technology Council, titled “Scientific Assessment of the Effects of Global Change on the United States,” included a section on “Attribution of Observed Climate Change to Human Activities at the Global and Continental Scale.” That section “addresses the extent to

which observed climate changes at the global and continental or national scale can be attributed to global anthropogenic emissions of greenhouse gases.” It states:

Difficulties remain in attributing temperature changes at smaller than continental scales and over time scales of less than 50 years. Attribution results at these scales have, with limited exceptions, not been established. Averaging over smaller regions reduces the natural variability less than does averaging over large regions, making it more difficult to distinguish between changes expected from external forcing and variability. In addition, temperature changes associated with some modes of variability are poorly simulated by models in some regions and seasons. Furthermore, the small-scale details of external forcing and the response simulated by models are less credible than large-scale features (IPCC, 2007a).

p. 70.

Until high-resolution computer models can predict with some degree of certainty future impacts of GHG emissions, CEQ should not require federal agencies to engage in such speculative exercises. **While it is appropriate for the draft guidance to refer to uncertainty and the limitations of regional modeling, the requirements of “downscaling and bias removal” for GCMs, and the need for agencies, in considering utilizing such modeling in their NEPA analysis, to disclose limitations, we question, in light of the above discussion, why agencies should undergo the burden and expenditure of resources of even considering such regional or local applications.**

In short, in light of their uncertainties and limitations, use of GCMs for regional and local NEPA analyses would be too speculative and unproductive, probably for some time to come, and we question CEQ’s inclusion of them in the draft guidance.

IV. Conclusion

In summary, EEI encourages CEQ to revisit in some fundamental ways the proposed guidance for consideration of GHGs and climate change in the NEPA context. While we agree that it often will be possible for federal agencies to estimate the anticipated GHG emissions of their proposed activities, CEQ needs to ensure that agencies in no way equate individual project GHG emissions, at or above 25,000 tons per year, as a “significant effect” warranting preparation of an EIS.

When compared with nationwide or global GHG emissions, 25,000 tons per year is too low a threshold to be meaningful for purposes of NEPA analysis. Indeed, the most helpful thing CEQ can and should do in its guidance on GHGs and climate change is to indicate that individual project GHG emissions typically will be minuscule compared to global emissions and so do not need to be studied in any qualitative way or substantial detail in the NEPA context.

Given the lack of any way to measure a causal link between individual project GHG emissions and specific climate change impacts, CEQ should ensure that the guidance does not require NEPA analyses to discuss the effect of an activity’s emissions on climate change, either quantitatively or qualitatively. Instead, the guidance should be limited to requiring publication of the activity’s projected annual emissions levels.

Finally, we question the ability of federal agencies to examine in a meaningful way the impact of climate change on a proposed activity – given that climate change modeling is fraught with

uncertainty, in particular when trying to ascertain climate change at a regional or local level – or with any degree of accuracy into the future. At most, agencies should be encouraged to

1) periodically check the evolution of climate change models and assess whether, in the future, these models have become sufficiently advanced so as to produce sound and robust regional and local estimates of specific environmental effects that may be useful to federal decision-makers, and 2) perhaps now only undertake a very simple and modest “scenario” analysis of potential impacts of climate change on their activity, to the extent that can be hypothesized, to see what aspects of the activity may need to be monitored as climate science evolves.